$\ln Y^P = \theta \ln K$

Critical Replication of Shaikh's Capacity Utilization Measure*

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May 21, 2026

as a directly measured slack capital

Abstract

Macroeconomic capacity utilization is conventionally constructed as a latent residual relative to an unobserved productive baseline. Shaikh (2016) reconstructs this baseline from a fitted long-run output–capital level relationship, centering his empirical analysis on cyclical residual dynamics rather than the structural properties of the underlying coefficients. This paper critically replicates that strategy by reinterpreting the estimated coefficient as a candidate transformation elasticity (θ) linking capital accumulation to productive-capacity formation. Operating within a progressive three-stage replication architecture, Stage S0 establishes that a Shaikh-like capacity path is arithmetically reproducible from corporate national accounts, though the baseline parameter remains sensitive to unstated data and lag choices. Stage S1 opens an expansive single-equation autoregressive distributed lag (ARDL) specification grid across 500 estimated models, revealing an admissible parameter space characterized by a disciplined non-uniqueness where the dynamic mean-reversion properties of utilization shift continuously with information-criterion penalty schedules. Finally, Stage S2 evaluates the relation within a multi-equation vector error correction model (VECM) framework, demonstrating that the standalone output–capital system fractures due to severe omitted variable bias. Systemic stability is achieved exclusively within a restricted, rank-one trivariate vector where the logged rate of exploitation explicitly anchors the cointegrating space. The paper proves that the output–capital multiplier cannot be interpreted as an insulated technical production law; instead, valid capacity identification must place distribution and institutional shock vectors directly inside the structural estimation architecture.

Keywords: Capacity Utilization; Critical Replication; Shaikh; Transformation Elasticity; Cointegration; ARDL; VECM; Distribution.
JEL Codes: B51; C32; E01; E22; E32; P16.

*This paper benefited from AI-assisted editing using Qwen3.6 Plus for syntax refinement, structural suggestions, and stylistic consistency checks. All analytical arguments, historical interpretations, and substantive content remain the author's sole responsibility.

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this paper, distinct from Shaikh

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1 Introduction

Capacity utilization sits at the center of macroeconomic debates in political economy, yet its structural measurement remains unresolved across both mainstream and heterodox traditions. This paper critically replicates the cointegration strategy developed by Shaikh (2016) for constructing a capacity utilization index, shifting the analytical focus away from residual variation and toward the structural parameter governing the output–capital relation. This underlying parameter is reinterpreted as the transformation elasticity of productive capacity with respect to the capital stock (θ). Placing the analytical weight on θ rather than on transient deviations from a fitted path allows Shaikh's empirical framework to be grounded within an explicit macroeconomic closure.

This structural reinterpretation introduces an unbalanced growth closure where the long-run transformation elasticity deviates from unity ($\theta \neq 1$). This formulation departs from the Harrodian knife-edge assumption of a unitary capacity transformation that routinely structures post-Keynesian and neoclassical modeling. Under an unbalanced growth closure, long-run disequilibrium operates as a persistent structural feature of accumulation rather than a temporary disturbance. Testing for stationarity then becomes an evaluation of whether aggregate demand buffers stabilize the expansion or whether capital accumulation follows a regime-dependent trajectory.

~~Executing this critical replication reveals that~~ utilizing the companion data tables to *Capitalism* ~~does not yield exactly reproducible~~ single-equation autoregressive distributed lag (ARDL) estimates. Approximate baseline results are recovered only after reconciling unstated data construction parameters and historical impulse controls. Furthermore, an open grid search across 500 admissible ARDL specifications demonstrates that the output–capital coefficient lacks unique identification. Preferred parameter estimates shift continuously with the information-criterion penalty schedules chosen by the researcher, exposing the fragility of single-equation cointegration claims.

I come close but do not fully replicate the original Sh result

Multi-equation vector error correction model (VECM) estimation exposes the definitive limits of this single-equation framework. When tested within a joint system framework, bivariate output–capital cointegration breaks down entirely across all estimated specifications due to severe omitted variable bias. Systemic stability is achieved exclusively within a restricted, rank-one trivariate system where the logged rate of exploitation enters the state vector. This system-level outcome demonstrates that the empirical tracking of the output–capital relationship is ~~fundamentally~~ conditioned by distributional variables and specific historical shock vectors.

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The broader objective is to evaluate whether the parameter that constructs the productive ceiling can sustain the structural meaning required by an unbalanced growth interpretation. Because productive capacity constitutes the denominator of the utilization index, any drift in the estimated coefficient mechanically shifts the historical level, variance, and mean-reverting properties of the recovered series. The empirical findings reveal that while the baseline benchmark is arithmetically reproducible, it remains parameter-sensitive within the single-equation grid and lacks self-sufficiency at the system level. Consequently, the identification of capitalist capacity paths cannot be resolved as a purely technical or engineering exercise.

a new variable is introduced & the estimates change

The article proceeds as follows. Section 2 ~~reconstructs~~ gives the history of the measurement problem ~~historically~~ with the, tracking how survey indicators and output-gap routines leave the productive ceiling theoretically underidentified. Section 3 establishes the conceptual framework, developing the properties of the transformation elasticity under balanced and unbalanced accumulation regimes. Section 4 implements the multi-stage empirical replication, detailing the baseline single-equation reconstruction, the multi-model ARDL sensitivity grid, the multi-equation VECM system tests, and the cross-stage econometric synthesis. Section 5 concludes by outlining the theoretical consequences for profitability

Can the coeffs on the new variable be meaningfully interpreted?

crisis models and positioning the endogenization of θ as a placeholder for future research.

analysis of the employment of existing physical capital

2 A Historical Trace of Capacity Utilization Measurement

This section reconstructs capacity-utilization measurement as a historical problem of statecraft and identification. Capacity utilization first became legible through survey-based ~~routines~~ tied to Fordist demand management, then was recoded into output-gap governance as macroeconomic institutions required portable benchmarks for slack. Political economy later reopened the question, ~~but did so through inherited measurement instruments whose construction limited what they could prove.~~ The section follows this historical trace across four steps: the emergence of survey-based capacity utilization as a practice of statecraft, its recoding into output-gap governance, the reopening of capacity utilization within political economy, and Shaikh's route from proxy measurement toward structural identification through profitability accounting.

say something affirmative — why?

— why let capital go unused?

2.1 From Survey Measures to an Inflation Sentinel: Capacity Utilization during the Fordist Era

Making idle productive capacity visible preceded its stabilization as a macroeconomic variable. The Great Depression rendered unused resources a political problem long before institutions possessed an administrative routine for measuring the productive ceiling. Brookings research converted scattered signs of downturn into a quantifiable threshold, requiring statisticians and policymakers to define capacity, specify approximation methods, and establish intervention boundaries. Early assessments quickly exposed the tension embedded in that conversion: engineering definitions of maximum throughput clashed with economic notions of sustainable operation, each carrying distinct implications for demand management and fiscal policy (Burns, 1954; Klein and David, 1958). From the outset, utilization measurement operated as statecraft through numbers rather than neutral statistical description.

citations

these are late for the Great Depression. Contrast developments in the measurement of unemployment

Hansen's secular stagnation thesis elevated productive under-utilization from a cyclical symptom to a structural threat to long-run growth, forcing stabilization policy to confront the productive ceiling directly (Hansen, 1936, 1941; Barber, 1987). Hicks and Samuelson codified this pressure into a pedagogical and analytical toolkit, embedding demand management within the IS–LM apparatus ~~that structured U.S. graduate training~~ (Hicks, 1937; Samuelson, 1947, 1948). Managing demand, employment, and bottlenecks required a metric that could translate idle capacity into intervention thresholds. Capacity utilization filled that role by rendering effective-demand shortfalls legible to a state apparatus committed to full-employment stabilization.

rework sentence

McGraw–Hill's plant-and-equipment surveys developed that role by regularizing managerial reports of actual and preferred operating rates. These surveys translated heterogeneous capacity definitions into a standardized reporting stream (Butler, 1958; Phillips, 1963). Institutionalizing the survey marginalized a deeper methodological debate over capacity measurement. It sedimented administrative routines for statistical production, making reproducibility the organizing principle for macroeconomic policy. Repeated collection generated temporal and sectoral comparability, while monopoly capital's information infrastructure anchored the measure's legitimacy in business forecasting and policy debate (Baran and Sweezy, 1988; Munroe, 2007). Firms, analysts, and policymakers coordinated around this administrative routine. It established a common language of slack without settling the technical measurement of productive ceilings.

this is kind of yadda yadda

* How did it relate to unemployment?

Like unemployment ?

The Federal Reserve incorporated survey-based capacity measures into the Industrial Production program, transforming the benchmark from one gauge among many into the standardized source for macroeconomic decision-making (Board of Governors of the Federal Reserve System, 2017). Institutional embedding raised the switching costs of alternatives and standardized what counted as admissible evidence. Physical-intensity measures such as Electric Motors Utilization and the Average Workweek of Capital tracked operational realities more directly (Foss, 1963; Taubman and Gottschalk, 1971), yet they never secured comparable bureaucratic adoption. Administrative path dependency, not empirical superiority, secured the survey routine's entry into the state's statistical machinery. — blah blah...

???

The stagflationary crisis of the 1970s ~~did not dismantle the survey routine~~; it recoded its policy function. Mounting inflation detached capacity utilization from its role as a demand-slack indicator and repositioned it as an inflation sentinel. ~~The benchmark survived, but its admissible meaning narrowed.~~ Where it once justified expansionary fiscal and monetary intervention, it now signaled when effective output threatened price stability. Brookings debates increasingly judged capacity measures by their ability to deliver credible inflation forecasts, subordinating employment diagnostics to price-level discipline (Klein et al., 1973). As the Bretton Woods order unraveled and profitability pressures intensified, the operational routine that had anchored full-employment management was repurposed for anti-inflation governance. This durability did not resolve the underlying identification problem. It cleared institutional space for a new governance architecture, one that would displace survey-centered utilization with output-gap benchmarks portable across central banks and supranational institutions. **

late 1960s gains + [illegible] ??

2.2 From Capacity Utilization to Output-Gap Governance

The erosion of Bretton Woods capital controls and the consolidation of cross-border payment networks dissolved the territorial anchor of Fordist measurement. Where survey-based capacity utilization had been embedded in national manufacturing sectors and tied to discretionary demand management, post-crisis governance required a benchmark capable of circulating across central banks, fiscal authorities, and supranational surveillance institutions (Jessop, 2002, 2003; Konings, 2007). The establishment of SWIFT, the acceleration of financial telecommunications, and the reorganization of production around global value chains generated a specific demand for indicators that operated independently of national industrial surveys. Real-time discretionary intervention yielded to rule-based frameworks whose authority rested on the consensus that depoliticized technocratic routines could substitute for political judgment (Saad-Filho, 2018). This displacement was not a technical upgrade. It reorganized macroeconomic statecraft around a portable architecture of slack.

??? wafty, defend or cut.

The Lucas critique formalized this rupture: discretionary policy became suspect precisely because it destabilized the expectations that credibility-based governance required to function (Johnson, 2024). Output-gap estimation operationalized that rupture, translating slack into a benchmark legible to central banks and supranational authorities. ~~Yet credibility was not discovered; it was enforced.~~ The Volcker Shock consolidated two decades of Eurodollar expansion and offshore intermediation into a new hegemony of financial accumulation (Konings, 2007; Krippner, 2005). Volcker's discretionary rate hikes were not rule-based; they operated as an act of monetary severity designed to establish the credibility that formal rules would later codify. What the 1980s enforced through demonstrated willingness to impose costs, the 1990s converted into portable architecture. The Taylor rule, inflation targeting, and central bank independence transformed disciplinary coercion into ~~a framework defensible as technically neutral~~ (Taylor, 1993; Johnson, 2024). Saad-Filho (2018)

why not just U. What does CU add?

** Simplify a lot. During the oil + price shocks of the 1970s, capacity utilization became one of the inflation predictors. → You're saying that the output gap was seen as key [illegible]

preference CU vs. unemp rate?

identifies this settlement as the New Monetary Policy Consensus: a policy regime that foreclosed the distributive contestation left open by demand management and recoded inflation control as the prerequisite for policy legitimacy. Fiscal constraint and the depoliticization of employment were rendered invisible by a framework that presented its own closure assumptions as the only arrangement consistent with stability. In Latin America, where structural adjustment programs compressed wage shares by an average of eight percentage points across the 1980s and 1990s, these closure costs were borne by populations long excluded from technocratic neutrality (Hickel et al., 2026).

Rise of output gap rhetoric

The output gap consolidated not because it solved the measurement problem, but because it could be produced through multiple, mutually reinforcing closure strategies. Actual output is recorded; potential output must be inferred. Each inference carries closure assumptions that fix the economy's trend and dictate the permissible distance between actual activity and capacity. Statistical filtering extracts trend from cycle through mechanical smoothing, treating potential as an emergent property of the series itself (Hodrick and Prescott, 1997; Baxter and King, 1999; Beveridge and Nelson, 1981; Hamilton, 2018). The production-function approach imposes neoclassical structure directly, deriving capacity from trend productivity, capital stock, and frictionally unemployed labor (Congressional Budget Office, 2004; Havik et al., 2014; Alichi, 2015). Structural VARs anchor potential to long-run restrictions, coding permanent movements as supply-driven and transitory deviations as demand shocks (Blanchard and Quah, 1989). Methodological divergence masks functional convergence. Each route collapses the politics of capacity identification into a single portable variable. Durability does not rest on empirical adequacy but on institutional versatility: the output gap operates as diagnostic in one context and enforcement tool in others. It sediments across the IMF, OECD, CBO, and EU fiscal architectures while its closure assumptions remain systematically unexamined in policy discourse.

X

closure ≠ policy variable

disconnect or drop.

The Global Financial Crisis did not dislodge this settlement; it exposed the conditions on which its durability depended. Post-crisis revisions routinely flipped the sign of the gap, inverting the policy signals the benchmark was designed to deliver (Kavřík, 2026). In the European periphery, structural-balance calculations built on downward-biased potential-output estimates by the European Commission compressed public investment during the deepest demand shortfalls in postwar history (Heimberger and Kapeller, 2017; Heimberger, 2020). Simultaneously, central banks faced intensifying pressure to calibrate unconventional monetary policy in real time, transforming the output gap from a retrospective diagnostic into a forward-looking intervention trigger. Institutional responses split along this fault line. Research divisions within supranational institutions documented hysteresis and revision volatility, showing that deep recessions permanently scar output and that real-time gap estimates carry errors large enough to reverse policy prescriptions (Cerra and Saxena, 2017; Cerra et al., 2023; Barkema et al., 2020). Operational arms, by contrast, accelerated measurement frequency and complexity. Nowcasting models, dynamic factor architectures, and forward-guidance frameworks converted the gap into a high-frequency signal for rate paths and asset purchases (Aastveit and Trovik, 2014; Berger et al., 2023). Patching the apparatus required discretionary interventions that steadily eroded the claim to rule-bound expertise on which its depoliticized authority rested (Foster, 2024; Argyroulis and Vagdoutis, 2025). Hence, the durability of measurement practices does not signal adequacy. Rather, it marks institutional entrenchment: a revision-prone, closure-dependent apparatus that now relies on the discretionary repair its original design was meant to eliminate. That is the setting in which political economy's capacity-utilization debates become historically intelligible.

– Really?

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2.3 Capacity-Utilization Debates in Political Economy

The utilization controversy divides along a structural fault line: whether the normal rate of capacity utilization is endogenous to demand or anchored in cost structures. The Neo-Kaleckian line treats the normal rate as demand-determined. Firms adjust desired utilization through expectation revision, micro-level shift-system responses to indivisible capital, or hysteresis that permanently resets the utilization target (Lavoie, 1995; Nikiforos, 2013; Setterfield and David-Avritzer, 2020). Accumulation trajectories lock into the demand regime that forms them, making distributive outcomes path-dependent. The Sraffian-Keynesian tradition anchors the normal rate in production costs and competitive conditions. Actual utilization gravitates toward this center over the long run, with convergence mediated by sluggish investment adjustment and reserve-capacity maintenance (Ciccone, 1986).

Both traditions face internal limits that block straightforward empirical resolution. Critics of the Neo-Kaleckian endogeneity argument question whether micro-level shift mechanisms aggregate coherently into a stable macro target. The Sraffian supermultiplier framework, which formalizes the convergence claim, confronts a deeper structural problem: it renders investment quasi-endogenous, passively adjusting to the growth rate of autonomous demand. Sustained autonomy for this demand component is also untenable under stock-flow consistency. Debt-financed expenditure generates liability accumulation that forces endogenous adjustment once balance sheets reach fragility thresholds, contradicting the model's closure assumption (Nikiforos, 2018). Empirical tests remain inconclusive. Cross-country series show short-run responsiveness to growth but negligible long-run correlation, a pattern interpreted by some as evidence of convergence and by others as statistical artifact rather than structural gravitation (Gahn and González, 2022).

distort. attr

The theoretical split structures competing accounts of long-run accumulation and the scope of demand management. The Sraffian-Keynesian tradition extends the principle of effective demand into the long run by allowing investment to adjust capacity toward a distributionally determined normal utilization rate. On this closure, demand shocks generate level effects while the system's growth trajectory remains anchored to autonomous expenditure. The Neo-Kaleckian alternative treats persistent demand shortfalls as structurally consequential, leaving accumulation paths and distributive outcomes path-dependent. This divergence is not merely about convergence or drift. It is a dispute over what the cycle fundamentally represents. Mainstream benchmarks treat fluctuations as inflation signals disciplined by the NAIRU and the New Keynesian Phillips curve (Friedman, 1968; Phelps, 1968; Galí, 2015); political economy reads them as distributional constraints shaped by reserve-army dynamics and accumulation trajectories (Goodwin, 1967; Marx, 1999). Because the two traditions isolate different cyclical objects, empirical tests cannot rely on benchmarks constructed to monitor inflation or manufacturing slack. Isolating long-run capacity behavior requires an instrument whose statistical properties register genuine structural movements rather than construction-induced mean reversion.

The Federal Reserve series cannot supply that instrument. Its statistical behavior reflects administrative design, not underlying economic dynamics. The construction process blends survey data on current operating rates with regression-based capacity estimates smoothed against capital stock and time trends, explicitly calibrated to approximate sustainable output under normal conditions rather than maximum technical capacity (Shapiro et al., 1989). Methodological opacity compounds the distortion: raw microdata, weighting schemes, and splicing protocols remain undisclosed, blocking independent reconstruction (Shaikh, 2016). The sectoral scope deepens the mismatch. Built to track industrial manufacturing during Fordist expansion, the series now measures

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distinction! see Pollin

amplify this material

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what's Haluska's argument?

a contracting sector while being deployed to characterize economy-wide accumulation dynamics. Inferring macroeconomic trajectories from a series confined to the industrial base confuses sectoral monitoring with structural analysis (Haluska, 2023). These design and scope constraints hardwire mean reversion into the series, rendering it incapable of registering chronic departures from normal utilization. Testing its stationarity therefore does not verify economic convergence; it verifies the Federal Reserve's own smoothing protocol (Nikiforos, 2016, 2021b). The controversy requires evidence across capital-depreciation horizons, roughly thirty years, but the series operates as a cyclical slack indicator optimized for short-run policy surveillance. Transferring a sectorally bounded, short-run monitoring tool into a long-run accumulation debate imposes a closure assumption that precludes the very test it is meant to perform.

The empirical debate has therefore shifted to proxy measures, but this has not closed the identification problem. Physical indicators like the average workweek of capital track machinery use rather than managerial assessments, and exhibit trending behavior where the FRB series remains stationary by construction (Nikiforos, 2016, 2021b). Neither single-series strategy nor single-equation identification recovers a structural relation jointly shaped by utilization, demand, accumulation, and capacity formation. If the long-run relationship is a system property, it cannot be identified through isolated proxies or cross-country stationarity tests. The proxy impasse requires a structural measurement route. Shaikh (Shaikh, 2016) shifts capacity estimation into profitability accounting. He reconstructs the capital stock through a generalized perpetual inventory method (GPIM) and derives the theoretical capacity path from the accounting link between actual and maximum profit rates at the normal cost of capital. Utilization is constructed as the residual deviation from this implied output–capital ratio. The accounting route gains analytical traction through data portability. Because it relies on standard national accounts for output and capital stock, it bypasses institutional survey constraints and applies across jurisdictions. This portability enables consistent series construction where physical proxies are unavailable and embeds the theoretical assumption that normal utilization adjusts to demand conditions rather than remaining fixed by technology. The residual construction establishes an accounting identity but leaves identification open. Defining utilization as a deterministic residual imposes a capacity path without verifying whether the underlying variables actually share a stochastic equilibrium. When identification remains weak, estimation becomes unstable and can register constructed residuals as false cointegration signals. The next stage operationalizes this route through a multivariate cointegrating framework that separates the accounting definition from the data-generating process, testing whether utilization, demand, accumulation, and capacity formation actually converge or drift.

compare normal rate

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how?

it must burden the problem

2.4 Shaikh's Structural Identification of Capacity Utilization

The proxy impasse requires a structural measurement route. (Shaikh, 2016) shifts capacity estimation into profitability accounting. He reconstructs the capital stock through a generalized perpetual inventory method (GPIM) and derives the theoretical capacity path from the accounting link between actual and maximum profit rates at the normal cost of capital. Utilization is constructed as the residual deviation from this implied output–capital ratio.

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exploits, no?

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3 Conceptual Framework

The structural critique of aggregate production functions establishes that constant algebraic accounting identities must not be conflated with behavioral or structural laws of production (Shaikh, 1974). Although Patomäki (2017) ~~outlines a plausible challenge to the closed-system~~ ontology of Shaikh (2016) due to supressing historical contingency, this critique targets philosophical appearance rather than analytical substance. The fragility in Shaikh's account does not stem from mathematical abstraction itself, but from a failure of model closure that simultaneously omits distributive conflict and enforces an artificial algebraic identity. While omitting these institutional distributive dynamics constitutes a profound specification vulnerability, resolving this omission remains outside the immediate scope of this essay. Instead, I focus strictly on this mismatch to demonstrate how Shaikh's single-equation baseline substitutes a statistical determinacy for a genuine behavioral closure, forcing the aggregate long-run output–capital coefficient to operate as a time-invariant technical law rather than a historically contingent transformation elasticity of productive capacities (Nikiforos, 2021a). see also Nelson & Plosser

This analytical limitation occurs because the framework fails to place the explanatory burden on the output–capital relation as a true transformation elasticity. Grounded in Marx's schemas of reproduction, smooth proportional growth represents a restrictive analytical abstraction rather than a historical description of accumulation. Relaxing this balanced-growth baseline highlights the presence of systematic instability, where a behavioral assumption with deep theoretical underpinnings is required to govern the long-run development of productive capacity. Shaikh's framework avoids this explicit structural parameterization, relying instead on the deterministic organization of an accounting identity. Consequently, the long-run coefficient remains a passive fitted slope that obscures the macroeconomic relation driving the transformation of accumulation into productive capacities.

Because the constructed capacity utilization series regulates the measurement of the normal rate of profit, this single-equation closure failure directly impacts the empirical findings of the classical surplus framework. Due to this direct functional dependency, the reconstructed utilization series inherits its dynamic properties from the foundational parameter, rendering subsequent conclusions regarding accumulation highly sensitive to the initial model choice. Instead of evaluating the statistical properties of the resulting residual, this section shifts the analytical focus toward the structural properties of this underlying coefficient. This section outlines these properties by isolating the functional dependency of the unobserved series, examining how the accounting identity mimics a production law, and contrasting the macro-regimes of balanced versus unbalanced growth closures.

Very longwinded

log $Y^P = a + bt$ is deterministic

Evaluating these conceptual boundaries requires an immediate, formal interrogation of the coefficient driving the productive-capacity path.

3.1 The coefficient driving the productive-capacity path

The empirical validity of a capacity-utilization series depends strictly on the stability of the parameter that generates the fitted capacity path. Shaikh (2016) avoids direct physical proxies, opting to reconstruct utilization from a capacity path embedded within an aggregate accounting identity:

where...

$$\ln Y_t = \ln K_t + \ln R^n_t + \ln \mu_t, \qquad (1)$$

define R_t

Within this specification, Y_t and K_t represent observed historical series from national accounts, while productive capacity Y^p_t and capacity utilization μ_t remain unobserved latent variables. While Shaikh (2016) utilizes a common price deflator to successfully eliminate inflationary distortions, this adjustment operates strictly within a monetary scale. Consequently, the specification retains a dimensional constraint because transcendental operations cannot carry constant-dollar units through the transformation, leaving the identity with uninterpretable unit structures like log-dollars (Basu, 2022a). Because logarithms evaluate underlying proportions rather than absolute levels, forcing unscaled monetary aggregates into the model renders the recovered parameters sensitive to the baseline unit of measurement.

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Isolating the long-run capacity path requires modeling the unobserved normal capacity–capital ratio ($\ln R^n_t$) as a linear function of a deterministic time trend and the observed capital stock:

$$\ln R^n_t = bt + c \ln K_t. \qquad (2)$$

Substituting this structural definition into the aggregate accounting identity yields the estimable single-equation level relation:

$$y_t = a + bt + d\, k_t + \varepsilon_t, \qquad d \equiv 1 + c, \qquad (3)$$

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Within this regression, y_t and k_t represent the natural logarithms of unscaled output and capital, while d represents the long-run output–capital elasticity. The ~~linear trend~~ component (bt) is conventionally interpreted as a measure of autonomous technical change. If the underlying long-run behavior is instead driven by a proportional growth relation ($g_{Y^p} = \theta g_K$), this level trend functions as an algebraic adjustment factor that absorbs the scale differences between the unscaled time series.

The fitted path of productive capacity follows directly from the recovered long-run components:

$$\hat{y}^p_t = \hat{a} + \hat{b}t + \hat{d}\, k_t. \qquad (4)$$

Capacity utilization is then constructed as the bounded deviation of realized output from this long-run capacity path:

$$\ln \hat{\mu}_t = y_t - \hat{y}^p_t, \qquad \hat{\mu}_t = \frac{Y_t}{\hat{Y}^p_t}. \qquad (5)$$

Because y_t and $\hat{y}^p_t$ share the same constant-dollar monetary units, this log-level subtraction algebraically collapses into the logarithm of a ratio. This operation allows the matching measurement units to cancel out completely, resolving the underlying scale mismatch and rendering the final extracted utilization series ($\hat{\mu}_t$) a dimensionally valid, unitless index (Basu, 2022a).

how and why?

This estimation sequence establishes a direct functional dependency, where the constructed utilization series inherits its level, timing, and variance from the estimated parameter $\hat{d}$. Consequently, different single-equation specifications yield divergent $\hat{d}$ estimates, producing contradictory trajectories for both capacity ~~and utilization~~. Because the model forces a dynamic accumulation process ~~into a rigid~~ accounting identity, the empirical sensitivity of this parameter raises an immediate identification problem. Structural measurement therefore rests on evaluating whether this long-run multiplier captures a genuine behavioral law of production, or whether it operates merely as an algebraic placeholder that absorbs omitted historical variation.

3.2 Laws of Algebra, Laws of Production, and Omitted Variable Bias

The estimated long-run coefficient d operates as the transformation elasticity ($\theta \equiv \partial \ln Y^p / \partial \ln K$) anchoring the profit-rate decomposition of Shaikh (2016). Because the normal rate of profit relies directly on the constructed capacity–capital ratio, the structural integrity of the entire accumulation theory depends on this specific parameter. While the aggregate accounting identity deterministically organizes these profit magnitudes, it leaves the underlying behavioral mechanism translating capital accumulation into productive capacity undefined. Following the analytical standard of Shaikh (1974), an accounting identity that leaves its core behavioral mechanism undefined mechanically mimics a structural production law, exposing the empirical framework to severe specification errors.

If the true long-run development of productive capacity operates as a proportional growth relation ($g_{Y^p} = \theta g_K$), the linear deterministic time trend inside the empirical specification functions as a mathematically superfluous level regressor. Simultaneously, the single-equation setup omits the historically shifting institutional vector, including distributive conflict and wage bargaining regimes. Omitting a relevant state vector while including an irrelevant deterministic trend generates a doubly misspecified model (Basu, 2020). Because the deterministic trend and the unobserved institutional shifts unfold contemporaneously across historical time, the trend acts as an accidental proxy. This correlation forces the estimated output–capital elasticity to absorb the omitted regime variation through an indirect bias channel.

A constant capital stock yields highly divergent capacity trajectories depending on specific historical regimes, including shifts in labor militancy, wage bargaining, and financialization. Driven by the indirect bias channel, the estimated single-equation elasticity absorbs this institutional variance, appearing stable only because it mathematically averages across these structurally heterogeneous historical mappings. Formally evaluating whether this recovered parameter isolates a genuine structural relation requires explicitly defining the theoretical conditions of balanced and unbalanced accumulation.

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3.3 Balanced and unbalanced growth

Marx's schemas of reproduction provide the analytical benchmark for proportional macroeconomic growth (Okishio, 2022). Smooth capitalist reproduction requires strict material proportions: commodities must sell, surplus value must be realized, and the material composition of output must match the input requirements of continuous accumulation (Basu, 2022b). Because real capitalist economies routinely violate these proportions, smooth reproduction remains an analytical abstraction rather than a historical description. To isolate the supply-side relation between capital and capacity, the long-run classical closure ($\mu = 1$) holds demand instability constant (Foley, 1985). Canonical post-Keynesian models enforce a balanced-growth baseline by assuming a unitary transformation

!!

OVB or misspecification

elasticity ($\theta = 1$) in the long run. Combined with demand fluctuations, this unitary assumption generates the characteristic properties of the Harrodian knife-edge.

Introducing an unbalanced growth closure ($\theta \neq 1$) achieves long-run identification by treating proportional accumulation as a restrictive special case. Relaxing the unitary assumption shifts the analytical focus away from the classical closure ($\mu = 1$). Under a non-unitary parameter, systematic instability is an inherent feature of capacity formation, distinct from temporary deviations driven by demand. The transformation elasticity (θ) governs the long-run development of productive capacity independently of short-run demand stabilization. In this setup, θ acts as a structural placeholder before its relation to distributive conflict is endogenized.

Defining the transformation elasticity as $\theta \equiv \partial \ln Y^p / \partial \ln K$, the analytical properties of the accumulation system depend on whether this parameter deviates from unity. Table 1 defines three macroeconomic regimes bounded by these conditions.

Table 1: Accumulation Regimes and the Productive-Capacity Relation

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Regime	Parameter Condition	Long-Run Relation ($\hat{Y}^p, \hat{K}$)	System Tendency
Balanced Growth	$\theta = 1.0$	Capital and capacity expand proportionally	Knife-edge baseline
Overaccumulation	$\theta < 1.0$	Capital expands faster than productive capacity	Accumulation decelerates toward stagnation
Excess Capacity	$\theta > 1.0$	Productive capacity expands faster than capital	Accumulation accelerates without supply-side limits

Note: This table defines the theoretical regimes governing the system.

These regimes represent the analytical limits of the accumulation process. Under an unbalanced growth closure without autonomous technical change, the dynamic path of the net capital growth rate ($\hat{k} \equiv \dot{K}/K - \delta$) is governed by the following ordinary differential equation:

whence (6)

$$\frac{d\hat{k}}{dt} = (\theta - 1)\, \hat{k}\, (\hat{k} + \delta), \tag{6}$$

where δ represents the depreciation rate. For any positive net accumulation ($\hat{k} > 0$), the sign of $d\hat{k}/dt$ depends on the magnitude of $(\theta - 1)$. Net accumulation decelerates under $\theta < 1$, accelerates under $\theta > 1$, and remains stationary under $\theta = 1$. This mathematical instability reflects the accounting laws of capital accumulation under a non-unitary transformation elasticity, distinct from Harrodian demand-driven dynamics.

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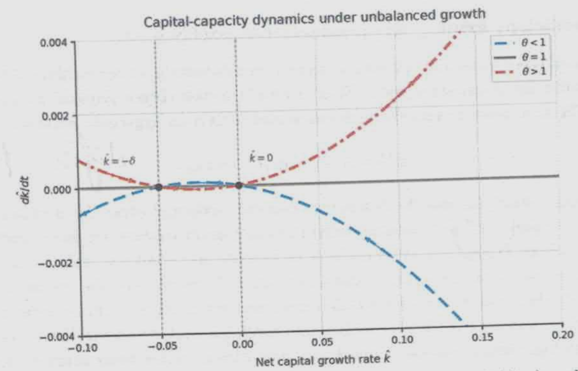
2nd derivative ??

Figure 1: Capital accumulation dynamics under unbalanced growth



Note: The phase diagram plots $d\hat{k}/dt$ against $\hat{k}$ for $\theta < 1$ (dashed line), $\theta = 1$ (solid line), and $\theta > 1$ (dash-dot line). Vertical dotted lines mark the equilibria at $\hat{k} = 0$ and $\hat{k} = -\delta$.

While effective demand and distributive conflict retain their analytical relevance, they are abstracted from in this initial formulation. Institutional and macroeconomic forces dampen, accelerate, or redirect the underlying long-run tendency but do not generate it. The differential dynamics in equation (6) imply a regime-dependent trajectory, yet the empirical single-equation model restricts θ to a time-invariant constant. Evaluating whether this parameter invariance holds historically establishes the exact structural implications of the fixed-parameter closure.

3.4 Trend stabilization and the fixed-parameter closure

Under the unbalanced growth closure ($g_{Y^p} = \theta g_K$), the dynamic trajectory lacks a feasible long-run equilibrium. As established in the phase dynamics, a system experiencing over-accumulation ($\theta < 1$) forces the net accumulation rate to decay to zero. To prevent this collapse, the empirical specification in Shaikh (2016) introduces a deterministic level trend (b), extending the productive-capacity closure ($g_{Y^p} = \theta g_K + b$).

Substituting this extended closure into the acceleration identity transforms the system into a quadratic ordinary differential equation:

$$\frac{d\hat{k}}{dt} = (\hat{k} + \delta)\left[(\theta - 1)\hat{k} + b\right]. \tag{7}$$

This specification shifts the equilibrium structure of the accumulation process. The stagnation limit ($\hat{k}^* = 0$) is replaced by a new interior root:

$$\hat{k}^* = \frac{b}{1 - \theta}. \tag{8}$$

problematic around θ≈1

Under over-accumulation ($\theta < 1$) with positive autonomous technical progress ($b > 0$), this equilibrium is locally stable. The depreciation rate (δ) anchors the negative root, while the ratio $b/(1-\theta)$ scales autonomous progress against the structural imbalance.

Stabilizing an unbalanced accumulation path through an exogenous algebraic term imposes a rigid theoretical restriction. The trend-stabilized closure forces a deterministic parameter (b) to offset structural imbalances, rather than endogenizing the institutional mechanisms regulating capacity formation. If technological change responds dynamically to class struggle and the wage share, the stabilizing attractor is a regime-dependent outcome of distributive conflict. Treating this algebraic stabilization as a behavioral law leaves the actual stabilization mechanism undefined (Shaikh, 1974).

lost

The empirical validity of the constructed utilization series relies strictly on establishing a valid cointegrating vector between output, capital, and this deterministic trend. Omitting distributive conflict while including the exogenous trend introduces a double misspecification (Basu, 2020). Because the trend mathematically absorbs unobserved institutional shifts, this indirect bias channel generates plausible misleading inferences regarding the existence and stability of the cointegrating relation itself. If the deterministic trend acts as an accidental proxy for regime changes, the fundamental statistical requirement of the single-equation specification is structurally compromised.

Estimating this relation over a multi-decade sample imposes a rigid temporal constraint. This temporal restriction forces both the elasticity (θ) and the trend (b) to act as time-invariant constants across shifting institutional environments. The classical long-run closure ($\mu = 1$) isolates theoretical capital-to-capacity relations (Foley, 1985) or smooth reproduction paths (Okishio, 2022; Basu, 2022b). Because real capitalist economies routinely violate these strict proportionality conditions, imposing time-invariance forces the recovered parameters to mathematically average across structurally heterogeneous historical periods.

??

The parameter θ acts as a theoretical placeholder. Section 4 implements an analytical stress test across three distinct stages to map the conditions under which this single-equation estimation fails. Stage S0 evaluates baseline recoverability using the exact historical data. Stage S1 maps the entire admissible specification space across a 500-model ARDL grid to evaluate parameter sensitivity to information-criterion penalty schedules. Stage S2 estimates the relationship using a multi-equation Vector Error Correction Model (VECM) to determine whether joint system stability requires the logged rate of exploitation ($e_t = \ln(\pi_t/(1-\pi_t))$) to enter the long-run cointegrating space. This sequence isolates the exact boundary conditions of the single-equation approach to preserve its portable features while establishing structural validity.

double ?

decept true + for what?

4 Critical Replication Study on Shaikh's Approach to Cointegration

Section 3 establishes that Shaikh's long-run output–capital coefficient is not a technical multiplier but the parameter that draws the productive-capacity path. Interpreted as the transformation elasticity θ, that coefficient links capital accumulation to capacity formation under unbalanced growth. The empirical question is whether ~~a fixed full-sample estimate sustains that production/reproduction meaning across postwar distributive regimes, or whether it averages over heterogeneous mappings.~~ This section implements a critical replication that tests the relation itself, not a utilization proxy.

The replication proceeds in three stages. Stage $S0$ reconstructs the closest recoverable version of Shaikh's single-equation procedure. Its purpose is narrow: to determine whether a capacity path reproduces from documented choices over data construction, deterministic closure, lag structure,

added variable

and historical impulse controls. Stage S1 opens the ARDL specification grid. It separates the bounds-tested admissible set from stricter confirmation criteria and information-complexity penalties, asking whether the recovered relation remains unique once admissibility and model-selection rules function as distinct researcher choices. Stage S2 moves from conditional single-equation estimation to joint system estimation. It tests whether the output–capital relation survives as a bivariate VECM and whether survival requires logged exploitation, $\ln e_t$, to enter the trivariate state vector.

The results narrow Shaikh's strategy ~~rather than reproduce it~~ mechanically. The single-equation coefficient reproduces, but it does not replicate the original specification exactly. Within the screened ARDL space, the retained region shifts once fit, parsimony, and complexity penalties separate. In joint-system estimation, the bilateral relation fails the admissibility gate; a restricted trivariate system survives only when exploitation enters explicitly. The coefficient interpreted as θ is therefore recoverable, non-unique within the single-equation grid, and system-admissible only under distributional conditioning.

The section reports these findings sequentially. Subsection 4.1 clarifies the identification strategy and shows how the long-run coefficient recovers from the national accounts. Subsection 4.2 details the data construction, measurement conventions, and treatment of structural breaks. Subsection 4.3 presents the staged replication design, moving from baseline recovery through ARDL specification searches to joint-system estimation. The objective is not to discard Shaikh's portability strategy, but to locate its boundary conditions. The replication demonstrates that the long-run output–capital coefficient identifies the capital–capacity mapping only when the historical organization of accumulation functions as a variable, not as a fixed backdrop.

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4.1 ARDL Estimation Architecture

Section 3.1 establishes that the long-run output–capital coefficient draws the productive-capacity path. This subsection operationalizes that coefficient within an autoregressive distributed lag framework, translating the accounting identity into a testable level relationship. The ARDL bounds-testing procedure (Pesaran et al., 2001) (PSS) evaluates the existence of a cointegrating relationship between output and capital regardless of their individual integration orders, eliminating pre-testing requirements for unit roots or cointegration rank.

The procedure estimates an unrestricted equilibrium correction model derived from a general ARDL(p, q) specification:

$$\Delta y_t = c_0 + c_1 t + \phi_1 y_{t-1} + \phi_2 k_{t-1} + \sum_{i=1}^{p-1} \gamma_i \Delta y_{t-i} + \sum_{j=0}^{q-1} \delta_j \Delta k_{t-j} + \sum_{h=1}^{H} \delta_h D_{h,t} + \varepsilon_t. \qquad (9)$$

Good

Equation (9) writes the ARDL specification in unrestricted error-correction form. Where c_0 represents the intercept, and $c_1 t$ the linear trend. The short-run dynamics is represented by Δy_{t-i} and Δk_{t-j}, and are included to capture the short-run memory of the model. The lagged level terms y_{t-1} and k_{t-1} carry the long-run content of the specification. The year dummmies block represent deterministic components that are excluded from the long-run relation $\sum_{h=1}^{H} \delta_h D_{h,t}$. This representation therefore separates short-run adjustment from the level relation between output and capital.

The asymptotic distribution of the F-statistic under H_0 depends explicitly on how deterministic constraints are imposed. Pesaran et al. (2001) identify five configurations (Cases I–V) for this constraint: Case I excludes both intercept and trend. Case II restricts the intercept to the

Give explicitly difference from Shaikh

cointegrating space. Case III includes an unrestricted intercept. Case IV includes an unrestricted intercept and restricts the trend coefficient to the null. Case V includes unrestricted intercept and trend. Each case shifts the critical value bounds because the null hypothesis alters the deterministic trending behavior of the level process. The bounds test rejects the null when the computed F-statistic exceeds the upper critical bound ($I(1)$), fails to reject when it falls below the lower bound ($I(0)$), and yields inconclusive inference when it lies within the interior band. The replication treats these deterministic configurations as admissible specification toggles. Systematic variation across Cases I–V, combined with lag-order selection and information-criterion penalties, defines the admissible specification grid.

The Wald F-statistic tests the joint exclusion restriction in equation (9). Under H_0^F, the lagged level terms do not enter the conditional model, so the specification contains no evidence of a long-run output–capital relation. Rejection of H_0^F suggests that y_{t-1} and k_{t-1} jointly contribute to the error-correction representation. The statistic is evaluated against the lower and upper bounds associated with an specific PSS deterministic case.

$$H_0^F : \phi_1 = \phi_2 = 0, \qquad H_a^F : (\phi_1, \phi_2) \neq (0, 0). \tag{10}$$

where ϕ_1 and ϕ_2 are the coefficients on the lagged level terms in the unrestricted error-correction representation. Under the null, the output–capital equation contains no long-run level relation. The distribution of the statistic is non-standard because the PSS procedure brackets the polar cases in which the regressors are purely $I(0)$ or purely $I(1)$. Values below the lower bound fail to reject the absence of a long-run relation; values above the upper bound support cointegration; values between the bounds remain inconclusive.

The t-bounds statistic provides a stricter diagnostic where the deterministic case makes it formally applicable. It tests

$$H_0^t : \phi_1 = 0, \qquad H_a^t : \phi_1 < 0. \tag{11}$$

Against the one-sided alternative $\phi_1 < 0$, where ϕ_1 is the coefficient on the lagged dependent variable in the conditional error-correction equation. Rejection supports an error-correction interpretation: deviations from the estimated long-run relation are followed by adjustment in subsequent periods. Because the sample is short, I evaluate the bounds evidence using the finite-sample critical values following Natsiopoulos and Tzeremes (2022), rather than relying only asymptotic PSS tables.

Shaikh (2016) operationalizes the ARDL framework to recover a single long-run output–capital multiplier, treating the coefficient as an accounting artifact rather than a production-theoretic parameter (Shaikh, 2016, Appendix 6.7). The baseline specification imposes four diagnostic constraints: (1) lag orders p and q follow the Akaike Information Criterion to mitigate residual serial correlation while avoiding over-parameterization; (2) deterministic treatment adopts PSS Case IV, retaining an unrestricted intercept while restricting the trend coefficient under the null; (3) year dummies (d_{56}, d_{74}, d_{80}) absorb outlier-driven residual spikes identified through squared-error diagnostics; (4) the Breusch–Godfrey LM test screens the ECM for autocorrelation before bounds inference proceeds. Where the F-statistic falls within the inconclusive interior band, the procedure defaults to Engle–Granger two-step estimation to establish integration order. This baseline architecture stabilizes the fitted capacity path but leaves the fixed-parameter closure assumption untested. Shaikh estimates d without theorizing its structural role or evaluating coefficient stability across distributive regimes. The replication treats this recovered multiplier as the empirical candidate for $\hat{\theta}$, routing the fixed-parameter closure assumption directly into three-stage admissibility gates.

has been used

Upon confirmation of a level relationship, the long-run coefficient recovers from the short-run ECM parameters:

$$\hat{d} = \frac{\sum_{j=0}^{q-1} \hat{\delta}_j}{1 - \sum_{i=1}^{p-1} \hat{\gamma}_i}. \tag{12}$$

In Shaikh's procedure, this long-run multiplier anchors the fitted capacity path. Thus, capacity utilization is constructed as the residual deviation from the long-run co-integration vector: $\log \hat{\mu}_t = y_t - \hat{y}_t^p$. This sequence establishes a strict dependency: utilization derives entirely from the estimated coefficient. Shaikh treats $\hat{d}$ as a fitted accounting multiplier; but as mentioned previously, he does not theorize its structural role. This replication re-interprets the coefficient as a transformation elasticity $\hat{\theta}$ of productive capacities, with respect to capital accumulation. Given the multiplicity of possible identification with the five cases of an ARDL model, and the year dummies used by Shaikh to control for outliers in historical episodes, the replication leverages on this researcher-decision making as sources of variation in the replication: conditionally on the specification grid, deterministic treatment, and admissibility gates. The replication stress-tests this assigned interpretation across three stages: reproducibility ($S0$), admissible-specification ($S1$), and system co-integration ($S2$). Subsection 4.2 details the data construction and measurement conventions required to execute these stages.

4.2 Data and Measurement

The replication uses annual US corporate sector data, 1947–2011 ($T \approx 65$), sourced from Shaikh (2016) data companions via Real Economic Analysis.[1] Capacity utilization is extracted as the bounded residual deviation from the cointegrated output–capital path; the fitted capacity path inherits its level, timing, and persistence from the measurement choices detailed below. Table 2 summarizes the core variables, their construction protocols, and the measurement constraints that structure the replication.

Note that this is already modeled

Capital stock follows the Generalized Perpetual Inventory Method (GPIM), an accounting framework that reconstructs *gross* capital stock from nominal gross investment flows, period depreciation rates, and implicit deflator built from nominal gross capital formation national accounts (Shaikh, 2016). The recursion accumulates current-cost capital as:

$$K_t = IG_t + z_t^* \cdot K_{t-1}, \qquad z_t^* \equiv (1 - z_t)\frac{p_t^K}{p_{t-1}^K}, \tag{13}$$

where IG_t is nominal gross investment, z_t is the period depreciation rate, and z_t^* reflates surviving capital to current replacement cost, or depletion rate of gross capital stock. Three adjustments anchor Shaikh's series: (1) BEA 1993 finite service lives replace the BEA 2011 infinite-lives assumption; (2) initial values are anchored to 1925 via BEA 1993 benchmarks; (3) the IRS corporate book-value index rescales the historical stock over 1925–1947 to account for Great Depression-era scrapping (Shaikh, 2016, Appendix 6.7, Sec. V). The resulting series tracks the BEA official net stock with a 99.60 % average ratio over 1947–2005, confirming that level shifts reflect methodological choices rather than estimation error. Full recursion inputs and depletion-rate construction are provided in Appendix B.4.

[1] Replication data and construction scripts are available at Real Economic Analysis: `https://www.realecon.org/data`. Shaikh (2016) specifies the common-deflator condition but does not name the price index; the replication implements this using a GPIM-consistent implicit price deflator to ensure reproducibility.

Table 2: Core replication variables: symbols, sources, and transformations

Symbol	Construction	Role	Measurement constraint
GVAcorp_t	NIPA Table 1.14 (L1) + imputed-interest adj. (Table 7.11)	Output flow	Classical surplus correction
KGCcorp_t	BEA Fixed Asset 6.1 (L9) + IRS SOI + GPIM recursion	Capital anchor	GPIM stock–flow reconstruction
p_t	pKN implicit price deflator, 2005=100	Common price basis	GPIM-consistent deflation
y_t	$\ln(\text{VAcorp}_t/p_t)$	Dependent variable	Inherits output and deflator choices
k_t	$\ln(\text{KGCcorp}_t/p_t)$	Regressor	Inherits GPIM depletion rates
D_{56}, D_{74}, D_{80}	Squared-residual diagnostics	Disturbance controls	Baseline outlier absorption

Note: Stage $S2$ introduces $\ln e_t$ (logged exploitation) to the trivariate state vector. Distributional variables enter only at the system-admissibility gate. Full series definitions, sources, and transformations are provided in Appendix 12.

Output and capital are deflated by the pKN implicit price index (2005=100), a GPIM-consistent implicit price deflator estimated under the same accumulation rules. Current BEA releases employ quality-adjusted chain weights that adjust for quality vintage, breaking stock-flow consistency for accumulation accounting. The pKN index preserves stock-flow consistency, ensuring the output–capital relation reflects real productive dynamics rather than quality-adjustment artifacts (Shaikh, 2016, Appendix 6.6, Sec. II). This satisfies the common-deflator condition specified in Shaikh (2016, Appendix 6.6, Eq. 6.6.7); the estimation rationale and comparison to quality-adjusted deflators are detailed in Appendix B.5.

Corporate gross value added is corrected for imputed financial intermediation services by subtracting net monetary interest paid and adding back imputed net interest adjustments (NIPA Table 7.11). The algebraic derivation and NIPA accounts usage considering their releases used by Shaikh, are detailed in more extension at Appendix B.2. Year dummies identified by Shaikh as residual-spike are used to as deterministic controls to absorb spikes given squared-error diagnostics (Patterson, 2000) in his ARDL baseline D_{56}, D_{74}, D_{80}; the author argues that they stabilize the initial specification by isolating outlier-driven deviations in 1956, 1974, and 1980. Historically, these set of dummies are coherent with relevant recessive events: the Eisenhower Recession, the Oil Shock crisis and the dismantle of the Keynesian State, and the Volcker shock of monetary policy. The sample size ($T \approx 65$) is small for time series inference, and is a source of potential inference bias on co-integration when relying on asymptotic critical value bounds which Shaikh does not consider in his original account; the replication addresses this by computing exact-sample bounds calibrated to $T = 65$ via stochastic simulation (Natsiopoulos and Tzeremes, 2022).

These measurement conventions constitute the identification strategy. The GPIM accounting approach to reconstruct gross capital stock, built the varaible of capital stock that is included in the analysis, the justification lies on the fact that net capital stock are representative of book-value and might be more appropriate to measure profitability, while gross capital stock by identifying

a depletion rate of capital in operation z_t^* that determines the path of accumualted gross capital stock. The assesment here is that Shaikh follows the adequate procedure to measure productive capacities with GPIM adjusted capital stock: they are more representative to measure capital stock in operation what is the interest when aiming to identify a measure related to productive capacities[2]; Shaikh also argue on the need to use a common-deflator that isolates real capital–capacity dynamics from price distortion effects.

4.3 Empirical Design and Admissibility Strategy

This section outlines the progressive multi-stage macroeconometric strategy used to evaluate the productive-capacity path. The critical replication begins with an exact baseline reconstruction in Stage S0. This initial stage directly tests the empirical reproducibility of the single-equation results reported by Shaikh (2016) before expanding the specification boundaries.

Secondly, the replication proceeds to stress-test the parameter sensitivity of Shaikh's single-equation model by turning arbitrary researcher choices and Autoregressive Distributed Lag (ARDL) structural properties into systematic sources of variation. In a nutshell, a combinatoric grid of 500 distinct ARDL specifications is constructed by permuting alternative dynamic lag lengths (p, q), historical outlier dummy configurations, and deterministic trend structures to explore the empirical distribution of the transformation elasticity (θ) and its underlying cointegration bounds properties. This stage encompasses an admissible-specification stress test organized across four distinct layers: (i) cointegration admissibility metrics evaluated through joint F-bounds and t-bounds statistics; (ii) specification invariance tracked via the variance of $\hat{\theta}$ across the model grid; (iii) research-driven identification fragility metrics; and (iv) capacity benchmark sensitivity evaluating the cascading effects of coefficient drift on the implied latent utilization series.

give simple explanat

Finally, the replication shifts from conditional single-equation models to a joint system framework using the Johansen (1991) reduced-rank approach. This system-level stage tests whether bivariate output–capital cointegration is robust to the canonical correlations of the data-generating process. By doing so, it evaluates whether introducing the logged rate of exploitation resolves structural underidentification by capturing explicit distributional conflict.

4.3.1 Stage 0 ($S0$): Reproducibility Test

Stage S0 tests the empirical reproducibility of Shaikh's results to establish a canonical point of departure for the replication and a benchmark for reconstruction. It selects Shaikh's published single-equation output–capital benchmark as the target, fixes the dynamic lag structure at the reported ARDL(2,4) configuration, and estimates the model using the closest recoverable corporate-sector series. This procedure systematically identifies the exact variables within the published data tables, safely bypassing undocumented documentation omissions noted in the original baseline text. Historical impulse variables and alternative Pesaran, Shin, and Smith (PSS) deterministic configurations are estimated to reconstruct the reported point estimates and establish a baseline, rather than to execute an open specification search.

The empirical procedure estimates the long-run output–capital relation, constructs productive capacity from that fitted path, and derives capacity utilization from the deviation of actual output

[2]Shaikh (2016) also makes the argument that gross capital stocks are also a preferable measure of capital stock to estimate profitability indices, this claim might be arguable from a material stand point, but capitalists decision making are not grounded on material reality, rather, on market valuation.

from the estimated capacity denominator. Cointegration bounds testing serves as the primary admissibility diagnostic to isolate valid level relationships from simple autoregressive dynamic filtering. This step highlights an essential technical property of the ARDL framework developed by Pesaran et al. (2001): while the F-bounds test remains universally applicable, the t-bounds statistic can only be formally defined under deterministic Cases 1, 3, and 5. Models tracking Case 2 or Case 4 configurations—which impose restricted constants or restricted trends within the cointegrating vector—cannot be subjected to a t-bounds confirmation gate.

4.3.2 Stage 1 ($S1$): Admissible-Specification Stress Test

The stage $S1$ protocol expands the conditional single-equation setup into an expansive multi-model space, generating a systematic grid of 500 estimated Autoregressive Distributed Lag (ARDL) specifications. By treating core modeling configurations as endogenous sources of structural variation, this framework evaluates how researcher choices over lag length, information criteria, and deterministic profiles affect cointegration bounds thresholds and long-run parameter identification. The resulting grid shifts the empirical focus from a singular baseline model to an entire admissible specification space, highlighting identification fragility under standard macroeconomic single-equation settings. This multidimensional grid strategy evaluates parameter sensitivity across changing information complexity indices, short-run structures, and historical outlier treatments.

To remove arbitrary lag choices, the grid lifts standard single-equation priors and permits full combinatoric variation across lag profiles such that $p, q \in \{1, 2, 3, 4, 5\}$. Every dynamic lag combination is estimated across all five deterministic cases outlined by Pesaran et al. (2001), paired with a nested block of four historical outlier controls: $s_0 = \varnothing$, $s_1 = \{D_{1974}\}$, $s_2 = \{D_{1974}, D_{1980}\}$, and $s_3 = \{D_{1956}, D_{1974}, D_{1980}\}$. These dummy frameworks absorb residual shocks linked to the Eisenhower recession, the 1974 oil shock, and the 1980 Volcker monetary contraction. This exhaustive permutation produces 500 raw single-equation models subject to structural admissibility gates. Formally, the complete specification space is defined as:

$$\mathcal{G}_{S1} = \{(p, q, c, s)\}, \tag{14}$$

where p and q represent the ARDL lag orders, $c \in \{1, \dots, 5\}$ indexes the deterministic configurations, and s indexes the nested baseline outlier structures.

The primary screen implements the F-bounds test to isolate valid level relationships from simple autoregressive dynamic filtering. Models qualify as admissible if their computed F-statistic rejects the null hypothesis of no long-run level relationship at a 90% confidence level using an asymptotic significance threshold, formally:

$$\mathcal{A}^F_{S1} = \{m \in \mathcal{G}_{S1} : p_F(m) \leq 0.10\}, \tag{15}$$

where $p_F(m)$ represents the asymptotic p-value from the joint F-bounds test for specification m. Specifications that generate F-statistics below the lower $I(0)$ bound or within the inconclusive interior band are systematically excluded from the F-admissible space $\mathcal{A}^F_{S1}$.

The secondary diagnostic screen introduces the t-bounds statistic to verify dynamic error-correction stability. Because t-bounds are mathematically defined only under PSS Cases 1, 3, and 5, specifications under restricted constant or restricted trend branches (Cases 2 and 4) bypass this

Is that too many exceptions?

layer without penalization:

$$\mathcal{T}_{S1} = \{m \in \mathcal{A}^F_{S1} : p_t(m) \leq 0.10 \text{ if defined}\}, \tag{16}$$

where $p_t(m)$ is the asymptotic p-value from the t-bounds diagnostic. The resulting $\mathcal{T}$-admissible space $\mathcal{T}_{S1}$ isolates specifications with robust evidence of a long-run relationship by overlapping the level-relation screen with an explicit test on the error-correction coefficient, rather than evaluating a joint test of the included variables. To identify high-confidence parameters, the Strong Cointegration Space ($\mathcal{C}_{S1}$) tightens the screening criteria to the 5% and 1% alpha thresholds:

$$\mathcal{C}_{S1} = \{m \in \mathcal{T}_{S1} : p_F(m) \leq 0.05 \text{ and } p_t(m) \leq 0.05\}, \tag{17}$$

which enforces robust empirical convergence properties across the remaining single-equation models.

Admissibility screens filter out invalid specifications but do not identify a unique parameter. Instead, the recovered parameters reflect a disciplined non-uniqueness that rejects the assumption of an invariant technical production law. Within the F-admissible pool, the first structural sorting layer constructs a Fit-Complexity Envelope (E_{S1}) to isolate optimal modeling trade-offs:

$$E_{S1} = \{m \in \mathcal{A}^F_{S1} : \nexists m' \in \mathcal{A}^F_{S1} \text{ with } L(m') \leq L(m) \text{ and } K(m') \leq K(m)\}, \tag{18}$$

where $L(m)$ is the lack of fit measured by $-2 \log L$, and $K(m)$ represents model complexity as measured by the total parameter count. This envelope isolates Pareto non-dominated specifications, mapping the geometric frontier where log-likelihood gains require expanding the parameter space.

The second sorting layer establishes information-criterion neighborhoods ($\mathcal{F}^{(0.20)}_j$) that act as "fattened" spaces around the geometric envelope. These neighborhoods relax strict optimizations to capture competitive models that sit slightly off the geometric Pareto frontier:

$$\mathcal{F}^{(0.20)}_j = \{m \in \mathcal{A}^F_{S1} : IC_j(m) \leq Q_{0.20}(IC_j \mid \mathcal{A}^F_{S1})\}, \tag{19}$$

where $j \in \{\text{AIC}, \text{BIC}, \text{HQ}, \text{ICOMP}, \text{RICOMP}\}$, and $Q_{0.20}$ denotes the 20th percentile cutoff with ties retained. Traditional scalar penalty schedules (AIC, BIC, HQ) are augmented by Bozdogan's information complexity indices, evaluating parameter covariance structures and inverse-Fisher geometric information properties. The RICOMP criterion actively screens the parsimonious edge of the envelope by penalizing parameter interdependence. Long-run parameter estimates are highly dependent on deterministic treatments and information-criterion penalty schedules, meaning different criteria locate divergent regions of the fit-complexity space.

Every specification m yields a distinct parameter estimate $\hat{\theta}(m)$, which directly shapes the fitted capacity path. The constructed utilization series then emerges as the stationary component of a flow-stock system, downstream from a distributionally sensitive parameter space:

$$\hat{\mu}_t(m) = y_t - [\hat{a}(m) + \hat{b}(m)t + \hat{\theta}(m)k_t]. \tag{20}$$

Tracking $\hat{\mu}_t(m)$ across $\mathcal{A}^F_{S1}$, E_{S1}, and $\mathcal{F}^{(0.20)}_j$ highlights the direct economic consequences of statistical parameter drift.

Table 3 details the operational layout of the $S1$ specification search. Rather than optimizing for a single candidate model, this framework establishes the empirical variance of the transformation parameter across defensible single-equation structures. This single-equation core mapping is evaluated empirically in Section 4.5.

Table 3: Synthesis of Admissibility Specification Stress Tests

Stress-Test Component	Operational Layer	Screening/Organization Rule	Diagnostic Target
Cointegration admissibility	F/T-bounds sequence	$p_F(m) \leq \alpha$, $p_t(m) \leq \alpha$ (if defined) for $\alpha \in \{0.10, 0.05, 0.01\}$	Existence and significance of long-run level relationships
Specification invariance	Lag-order variation	Monitor parameter drift $\hat{\theta}(m)$ across combinatoric $p, q \in \{1, \ldots, 5\}$	Stability and unique mapping of the long-run structural multiplier
Identification fragility	Deterministic cases & dummy structures	Track admissibility shifts across $c \in \{1, \ldots, 5\}$ and $s \in \{s_0, \ldots, s_3\}$	Sensitivity to deterministic components and structural trend assumptions
Information complexity & Pareto optimization	Fit-complexity envelope & IC neighborhoods	Pareto non-dominance in E_{S1}; 20th percentile cutoff for $j \in \{\text{AIC, BIC, HQ, ICOMP, RICOMP}\}$	Organization of disciplined non-uniqueness under distinct information schedules
Capacity benchmark sensitivity	Utilization path fan $\hat{\mu}_t(m)$	Map structural parameter variance to latent residual series	Macroeconomic consequence of single-equation coefficient drift on normal profit estimation

Note: This synthesis matrix outlines the nested single-equation admissibility screens, information-theoretic optimization layers, and diagnostic targets within the stage $S1$ multi-model grid. Source: Author's elaboration.

Section 4.3.3 extends the critique to a joint multi-equation system. Moving beyond single-equation constraints, the system-level analysis tests whether bivariate cointegration fractures due to severe omitted variable bias (OVB). This structural breakdown dictates whether joint system survival requires the logged rate of exploitation (e_t) to enter the long-run cointegrating space.

4.3.3 Stage 2 ($S2$) System Co-Integration Test

Stage S2 evaluates whether the bilateral output–capital relation remains self-sufficient when shifted from a conditional single-equation setup to a joint multi-equation system. The baseline bivariate vector, $X_t = (\ln Y_t, \ln K_t)'$, immediately fractures due to severe omitted variable bias (OVB), failing the fundamental Johansen rank conditions and companion-matrix stability gates. To resolve this critical underidentification, the expanded framework introduces explicit distributional conditioning by integrating the logged rate of exploitation, $e_t = \ln(\pi_t/(1-\pi_t))$, yielding the trivariate state vector $X_t = (\ln Y_t, \ln K_t, \ln e_t)'$. This trivariate system is estimated using the full Johansen (1991) Vector Error Correction Model (VECM) representation:

$$\Delta X_t = \Pi X_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta X_{t-i} + \Phi D_t + \varepsilon_t, \tag{21}$$

good – hold up
against the Shaikh version

where Γ_i captures short-run dynamics, ΦD_t contains deterministic controls, and the long-run impact matrix is factored as $\Pi = \alpha\beta'$ to evaluate reduced-rank stability. Joint system survival is achieved exclusively under this formulation, where the rank condition $r = 1$ holds and the exploitation variable successfully anchors the long-run cointegrating space (β).

To systematically test this stability, S2 implements a VECM combinatoric grid mirroring the single-equation sensitivity approach. Formally, the complete attempted specification space for each target rank (r) is defined as:

$$\mathcal{G}_{S2}^{(r)} = \{(p, c, s)\}, \tag{22}$$

where $p \in \{1, 2, 3, 4\}$ represents the VAR lag orders, $c \in \{d_0, d_1, d_2, d_3\}$ indexes the Johansen deterministic closure configurations, and $s \in \{h_0, h_1, h_2\}$ isolates a subset of the baseline historical outlier controls. Permuting across these state-vectors yields an attempted grid of 48 specifications per rank. However, because specifications under the restricted constant branch (d_1) systematically fail numerical convergence across all permutations, the successfully estimated baseline is analytically restricted to exactly 36 models per system family.

The Stage S2 admissibility protocol evaluates these surviving 36 specifications through a rigorous triple gate requiring numerical convergence, Johansen rank confirmation, and companion-matrix dynamic stability. Within the trivariate framework, the reduced-rank restriction $r = 1$ tests whether a single cointegrating relation survives distributional integration, while $r = 2$ evaluates the potential for a dual-relation system. To organize the surviving specifications, the subset Ω_{20} ranks the admissible models based strictly on minimized log-likelihood loss. This likelihood-based organization layer establishes an interpretive hierarchy for the retained systems but does not independently define the primary boundaries of system admissibility.

4.3.4 Cross-Stage Synthesis

The multi-stage critical replication executes an intentionally nested screening architecture to map the boundary conditions of the productive-capacity path. Stage S0 establishes the approximate empirical recoverability of the baseline single-equation framework. Stage S1 expands the estimation framework into a comprehensive specification grid, utilizing structural bounds diagnostics and information-criterion envelopes to map parameter non-uniqueness. Finally, Stage S2 imposes multi-equation Vector Error Correction Model (VECM) stability gates to verify whether the underlying structural relation survives when estimated jointly. Table 11 formalizes this progressive methodological sequence.

4.4 Stage 0 ($S0$): Single-Equation Reconstruction

Stage S0 tests the empirical recoverability of the published fitted capacity path from the nearest reproducible single-equation output–capital specification. This initial stage establishes a canonical point of departure for the replication, testing point-estimate recoverability rather than parameter uniqueness or system-level survival. Shaikh (2016) estimates d, the long-run output–capital coefficient. This article interprets the empirical estimate $\hat{d}$ as $\hat{\theta}$ strictly because Section 3 assigns θ a theoretical role as the candidate transformation elasticity linking capital accumulation to productive-capacity formation.

The exercise represents a controlled reconstruction test rather than a mechanical reproduction. Because Shaikh (2016) omits a complete reporting trail of result-level estimates and variable provenance, S0 explicitly defines the baseline choices governing data construction, deterministic

Table 4: Stage Design and Admissibility Strategy

Stage	Design Object	Specification Space	Admissibility and Organization	Interpretive Implication
S0	Single-point reconstruction of baseline specification	ARDL(2,4), historical impulse dummies, alternative deterministic closures	F-bounds primary screen; t-bounds diagnostic where defined; utilization path proximity	Tests whether a deterministic capacity path can be closely reconstructed
S1	ARDL specification grid $\mathcal{G}_{S1}$	Lag orders (p, q), five PSS cases, four nested dummy structures	F-admissible set $\mathcal{A}^F_{S1}$; F+t confirmed candidates $\mathcal{C}_{S1}$; corrected envelope E_{S1}; IC-specific neighborhoods $\mathcal{F}_j^{(0.20)}$	Tests whether the output–capital coefficient is structurally unique or varies with IC penalty logic
S2 Bivariate	Joint $(\ln Y_t, \ln K_t)'$ VECM	Lag order, deterministic branch, dummy structure, $r = 1$	Triple gate $\mathcal{A}_{S2}$: convergence, Johansen rank, companion stability	Tests whether the bilateral output–capital relation is self-sufficient at the system level
S2 Trivariate	Joint $(\ln Y_t, \ln K_t, \ln e_t)'$ VECM	Same as bivariate, with logged rate of exploitation included; $r = 1$ or $r = 2$	Same triple gate; Ω_{20} organizes admissible survivors by log-likelihood loss	Tests whether the structural relation survives exclusively when distribution enters explicitly

Note: This table outlines the progressive multi-stage replication architecture. The sequence advances from baseline single-equation reconstruction (S0) through parameter sensitivity grids (S1) to joint system-level survival testing (S2). Source: Author's elaboration.

treatment, and long-run closure. The result establishes the closest recoverable benchmark under these documented historical impulse controls and lag structures.

The published reference value is $\hat{\theta} = 0.66$. The faithful reconstruction, which also emerges as the BIC winner in the S0 rerun, is an ARDL(2,4) specification yielding $\hat{\theta} = 0.72$. The AIC comparator selects an ARDL(4,3) specification and yields $\hat{\theta} = 0.75$. Table 5 reports the precise parameters for these two estimated S0 long-run multipliers. The wedge between 0.66 and 0.72 constitutes the primary S0 result: the benchmark is closely recoverable from the national accounts, but it is not exactly reproduced.

Table 5: Stage S0 Long-Run Output–Capital Multipliers

Reference/Model	Order	Role	$\hat{\theta}$	SE	t	p
Shaikh (2016) Benchmark	ARDL(2,4)	Published reference	0.66	–	–	–
Faithful Reconstruction	ARDL(2,4)	Reconstructed benchmark	0.72	0.06	11.82	< 0.001
AIC Comparator	ARDL(4,3)	Lag-length comparator	0.75	0.04	18.43	< 0.001

Note: This table reports Stage S0 long-run output–capital multipliers. Shaikh's published estimate operates as the baseline reference value; standard errors, t-statistics, and p-values are reported exclusively for the generated replication estimates. Source: Author's elaboration.

Coefficient proximity is not structurally equivalent to cointegration admissibility. Consequently, S0 evaluates the reconstructed coefficient alongside formal ARDL bounds evidence. The F-bounds statistic supplies the primary admissibility gate for a level relation, utilizing small-sample ARDL bounds inference and case-specific critical values. The t-bounds statistic operates as a stricter

diagnostic where the deterministic case formally defines it; this secondary statistic tightens the pool but does not replace the primary F-bounds gate.

Table 6: Stage S0 ARDL Bounds Evidence at $\alpha = 0.10$ (90% Confidence Intervals)

Model	Test	PSS Case	Statistic	I(0) Bound	I(1) Bound	Exact-Sample p	Verdict
ARDL(2,4)	F-bounds	Case 1	3.349	2.466	3.333	0.099	Reject H_0
ARDL(2,4)	t-bounds	Case 1	-2.507	-1.602	-2.259	0.062	Reject H_0
ARDL(4,3)	F-bounds	Case 1	5.250	2.486	3.338	0.023	Reject H_0
ARDL(4,3)	t-bounds	Case 1	-3.125	-1.621	-2.270	0.015	Reject H_0

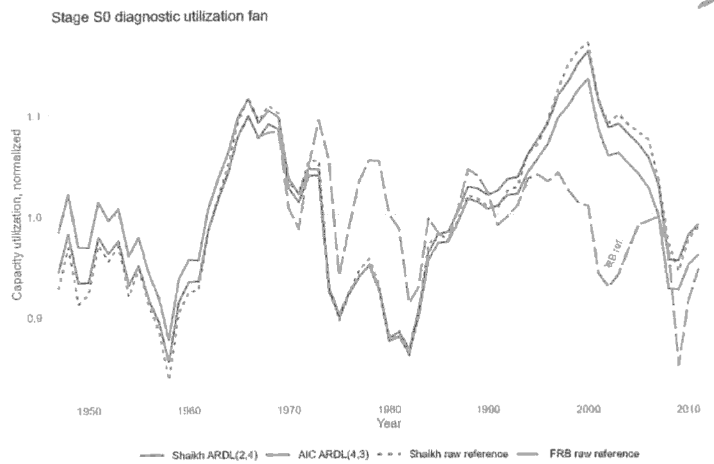
Note: The null hypothesis assumes no long-run level relation. The F-bounds statistic supplies the primary admissibility diagnostic. The t-bounds statistic is reported as a stricter secondary diagnostic where the deterministic structure permits. Case 1 denotes the no-constant/no-trend PSS deterministic configuration. Rejection is assessed at $\alpha = 0.10$. Source: Author's elaboration.

Figure 2 provides the primary path-level evidence for S0. The visualization maps a family of reconstructed utilization paths generated by the faithful reconstruction, alternating deterministic closures, and comparator specifications. These paths diverge because the constructed utilization series acts as the stationary component of a flow-stock system, downstream from a distributionally sensitive capacity parameter. Consequently, any drift in the long-run output–capital coefficient mathematically alters the implied utilization trajectory. Raw Shaikh and Federal Reserve utilization series are overlaid strictly as historical reference paths rather than empirical targets.

I have very little idea of what is going on here. Simplify and explain

Interpret economically

Figure 2: S0 Diagnostic Utilization Fan



Note: The fan maps the variance of constructed utilization paths given competing ARDL specifications. Source: Author's elaboration.

Stage S0 confirms that a Shaikh-like capacity benchmark is empirically reproducible. However, recovering a single point estimate does not guarantee that the underlying structural parameter is unique or robust. To expose potential identification fragility, Stage S1 expands the estimation framework. This next stage tests whether the recovered benchmark remains stable across a wide grid of admissible single-equation alternatives.

4.5 Stage 1 ($S1$) Admissible-Specification Stress Test Results

Stage S1 fixes the underlying data layer and opens the single-equation ARDL specification grid. This stage tests whether the recoverable baseline benchmark remains structurally unique once lag order, deterministic case, dummy structure, and information criteria systematically vary. In the formal notation of Section 4.3, this framework pivots from a singular ARDL point estimate to the fully screened grid space $\mathcal{G}_{S1}$.

The resulting specification grid is populated but does not collapse around a single parameter. The full multidimensional grid contains 500 combinatorial specifications, all of which estimate successfully without numerical failure. Applying the initial 10 percent bounds threshold, 102 specifications successfully pass the primary F-bounds gate and enter the F-admissible set $\mathcal{A}^{F}_{S1}$. Tightening the asymptotic significance size predictably reduces this retained set: at the 5 percent threshold, 62 specifications remain F-admissible, shrinking to just 13 valid specifications at the strict 1 percent limit.

Within this F-admissible pool, the secondary t-bounds diagnostic enforces a stricter requirement

for dynamic error-correction stability. At the 10 percent level, 49 specifications pass this confirmed-candidate layer $\mathcal{C}^{F,t}_{S1}$, representing cases where the t-bounds statistic is both mathematically defined and statistically significant. At the 5 percent level, 28 specifications achieve this dual confirmation, dropping to only 3 confirmed specifications at the 1 percent boundary. The t-bounds diagnostic records stricter confirmation where available, but it does not technically override or replace the primary F-bounds cointegration gate.

Table 7: S1 Shrinking Specification Space Counts

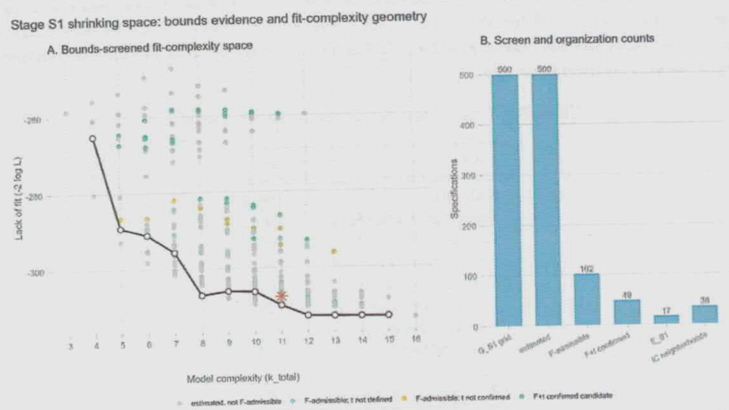
Layer	Count
Full grid	500
Estimated successfully	500
F-admissible at 10%	102
F+t confirmed at 10%	49
F-admissible at 5%	62
F+t confirmed at 5%	28
F-admissible at 1%	13
F+t confirmed at 1%	3
E_{S1}	17
IC-neighborhood union	36

Note: This table tracks the survival rate of the 500 estimated single-equation ARDL models through the nested cointegration admissibility gates. Source: Author's elaboration.

Figure 3 reports this screening hierarchy visually. Panel A maps the 500 estimated specifications across the fit–complexity space, color-coded strictly by bounds-admissibility status. Panel B translates this geometry into exact specification counts surviving each sequential screening layer. This sequential ordering reinforces the fundamental methodological rule that model comparison begins strictly after the bounds screen, not before it.

What is k?

Figure 3: S1 Shrinking Specification Space



Source: Author's elaboration.

Within the F-admissible set, the corrected global non-dominated fit–complexity envelope (E_{S1}) isolates 17 optimal specifications. The horizontal dimension of this geometric space measures model complexity via the total number of estimated parameters, while the vertical dimension measures the lack of fit, reported formally as $-2\log L$. Moving along the E_{S1} frontier therefore trades statistical fit against structural parsimony. The resulting envelope serves strictly as a geometric summary of Pareto-optimal ARDL alternatives, not as a singular new estimator.

The IC-specific neighborhoods operate as post-admissibility, criterion-conditioned organization layers. For each criterion $j \in \{\text{AIC}, \text{BIC}, \text{HQ}, \text{ICOMP}, \text{RICOMP}\}$, the neighborhood $\mathcal{F}_j^{(0.20)}$ isolates the bottom 20 percent of F-admissible specifications, capturing 36 unique models across the union. Table 8 reports the precise parameter ranges across these respective criterion borders, demonstrating that these neighborhoods organize the retained specifications rather than establish cointegration natively.

? What is a specification and what is better?

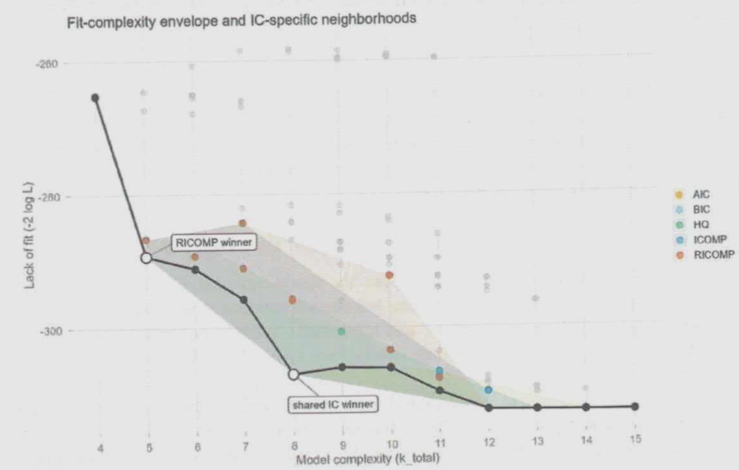
Table 8: S1 IC-Specific Neighborhood Summaries

Criterion	Specs	Min $\hat{\theta}$	Mean $\hat{\theta}$	Median $\hat{\theta}$	Max $\hat{\theta}$
AIC	22	0.84	0.88	0.86	0.93
BIC	21	0.65	0.91	0.86	2.20
HQ	21	0.75	0.94	0.86	2.20
ICOMP	22	0.65	0.84	0.86	0.95
RICOMP	22	0.65	0.83	0.85	0.95

Note: The table reports the summary statistics for the long-run parameter $\hat{\theta}$ within the bottom 20th percentile of each information criterion. Estimates are rounded to two decimal places for consistency. Source: Author's elaboration.

Figure 4 supplies the primary fit–complexity and IC-neighborhood evidence. The solid black line denotes the Pareto non-dominated fit–complexity envelope (E_{S1}). Surrounding this frontier, the shaded polygons define the 20th percentile neighborhoods for each respective information criterion. This visualization maps the spatial divergence between the highly parsimonious RICOMP winner and the shared scalar-penalty winner.

Figure 4: S1 Fit–Complexity Envelope and IC-Specific Neighborhoods

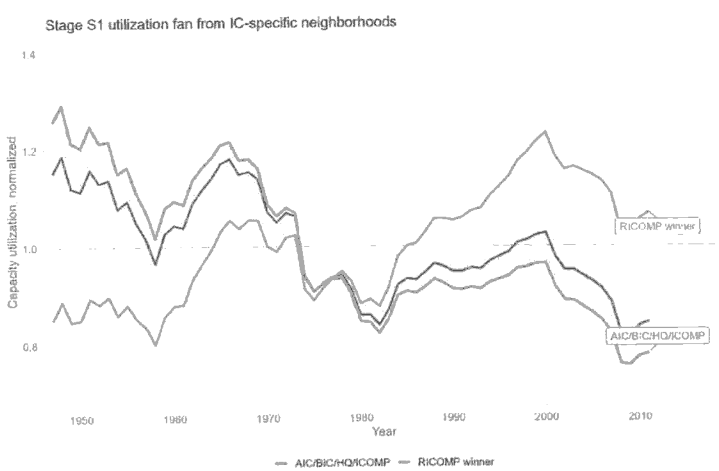


Source: Author's elaboration.

The shared AIC/BIC/HQ/ICOMP winner is an ARDL(3,3), Case 1 specification with the 1974 historical dummy (s_1), yielding $\hat{\theta} = 0.92$. Conversely, RICOMP selects a highly parsimonious ARDL(1,2), Case 2 specification with no impulse dummies (s_0), yielding $\hat{\theta} = 0.65$. This criteria divergence does not establish a structural ranking of the underlying estimators. Unlike standard scalar penalties, the ICOMP and RICOMP criteria evaluate complexity through the covariance structure of estimated parameters and residual dependence (Bozdogan, 1990, 2000). Consequently, the RICOMP neighborhood marks a covariance-complexity-sensitive region of the admissible space rather than providing a definitive model-selection verdict (Güney et al., 2021).

Figure 5 serves a narrower evidentiary role by reporting the diagnostic constructed-utilization paths generated strictly from the retained IC-neighborhood specifications. The shaded region maps the variance of constructed capacity utilization, while the divergent paths of the distinct criteria winners highlight the macroeconomic sensitivity to single-equation penalty schedules. These paths provide supporting evidence for parameter non-uniqueness, demonstrating how the implied utilization series mechanically shifts when the capacity denominator relies on different retained ARDL alternatives.

Figure 5: S1 Diagnostic Utilization Fan from IC-Specific Neighborhoods



Source: Author's elaboration.

Stage S1 demonstrates that the admissible single-equation space is populated, yet it systematically rejects the assumption of a unique structural benchmark. Because the retained alternatives fail to collapse into a single transformation elasticity ($\hat{\theta}$), the resulting single-equation utilization path remains structurally fragile. Section 4.3.3 addresses this fragility by testing whether the output-capital relation survives multi-equation system estimation, and whether that survival strictly requires distributional variables to enter the state vector.

Are you saying it's a critical test?

4.6 Stage 2 ($S2$) System-Level VECM Replication

Stage S2 shifts the empirical analysis from conditional single-equation specifications to a joint system framework. This multi-equation estimation evaluates whether the output–capital relationship maintains structural integrity once the fields of accumulation, capacity utilization, and distribution are treated as mutually endogenous. By transitioning to a vector error correction framework, the replication tests whether the long-run classical-Marxian production relation survives at the canonical correlations of the macroeconomic data. ????

The baseline bivariate state vector is restricted to output and capital, defined as $X_t = (\ln Y_t, \ln K_t)'$. To address potential omitted variable bias stemming from class conflict and structural profitability shifts, the expanded trivariate vector incorporates a classical measure of distribution. Raw exploitation is constructed as the aggregate corporate profit-share-to-wage-share ratio, $e_t = \pi_t/(1-\pi_t)$. The system-level estimation integrates this relationship using the logged rate of exploitation, $X_t = (\ln Y_t, \ln K_t, \ln e_t)'$, ensuring that distributional variables operate inside the cointegrating space rather than as downstream exogenous filters. ??

Systemic admissibility is governed by a triple confirmation gate requiring numerical convergence, Johansen reduced-rank verification, and companion-matrix dynamic stability. This protocol isolates robust long-run systems from unstable vector autoregressive processes. Treating the rank and deterministic choices as endogenous attributes prevents the estimation from mistaking simple mathematical convergence for an underlying macroeconomic equilibrium.

Table 9 details the structural outcomes across the complete VECM specification space ($\mathcal{G}_{S2}$) through the sequential gates of convergence, rank, and companion-matrix stability. The bivariate $r = 1$ configuration yields zero admissible models across all 36 estimated permutations, signaling a severe breakdown in the self-sufficiency of the output–capital relation. Conversely, the trivariate $r = 1$ framework yields six admissible specifications when conditioned on the logged rate of exploitation. The trivariate rank-two permutations confirm this restriction by producing zero stable dual-vector systems.

Table 9: S2 System Admissibility Outcomes

System	Rank	Estimated	Admissible	Interpretation
Bivariate $(\ln Y, \ln K)$	$r = 1$	36	0	Output–capital relation is not self-sufficient
Trivariate $(\ln Y, \ln K, \ln e)$	$r = 1$	36	6	Restricted survival with logged exploitation
Trivariate $(\ln Y, \ln K, \ln e)$	$r = 2$	36	0	No admissible two-vector system

Source: Author's elaboration.

Table 10 uncovers the parameter architecture of the six surviving trivariate models. Admissibility is strictly conditional on the h_2 historical dummy vector, which accounts for the step shifts of 1956, 1974, and 1980. This uniform survival pattern implies that Shaikh's single-equation outlier controls capture essential structural breaks required for system-level stability. While all six specifications validate the single-rank restriction ($r = 1$), the long-run cointegrating coefficients ($\hat{\theta}$) exhibit significant variation across different dynamic lag structures and deterministic treatments. Admissible specifications under the d_3 trend branch produce explosive or negative elasticities, meaning that economically viable long-run parameters lack viability and are constrained strictly to the d_0 and d_2 deterministic closures.

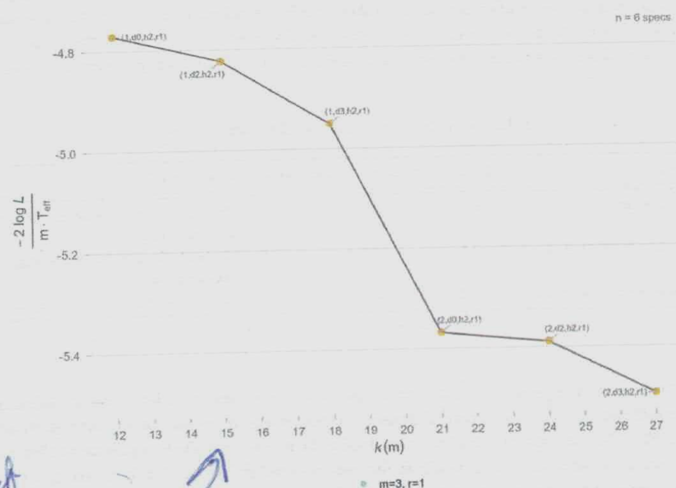
Table 10: S2 Retained Trivariate Specifications

Lag Order (p)	Deterministic Branch (d)	Historical Dummy	Rank (r)	Cointegrating Coefficient ($\hat{\theta}$)
1	d_0	h_2	1	0.91
1	d_2	h_2	1	0.97
1	d_3	h_2	1	-0.81
2	d_0	h_2	1	1.19
2	d_2	h_2	1	0.73
2	d_3	h_2	1	11.31

Source: Author's elaboration.

Figure 6 charts the fit–complexity properties of the surviving models. The empirical frontier confirms that system survival is restricted to a narrow parameter channel along the trivariate rank-one branch. Bivariate configurations cannot clear the multi-equation stability gate, and expanding the state vector to a rank-two setup fails to support a broader multi-vector interpretation of the productive relation.

Figure 6: S2 Pooled System Frontier



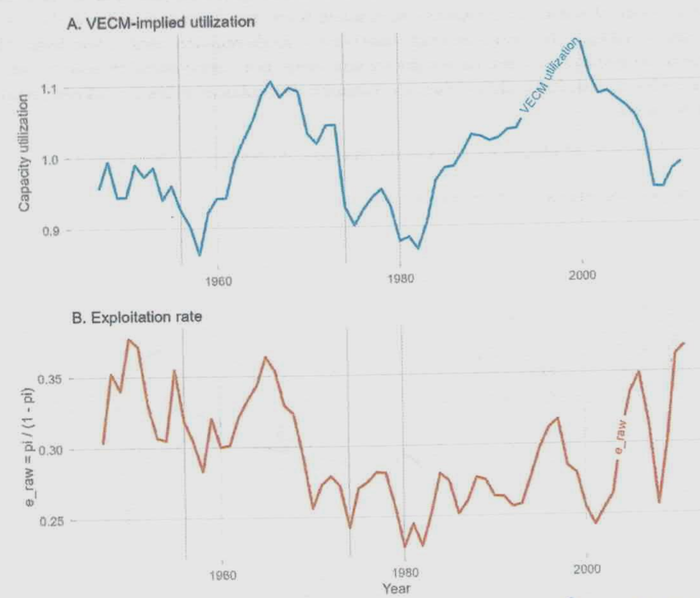
What is this?

Source: Author's elaboration.

Figure 7 records the dynamics of the final system vector. Panel A outlines the VECM-implied

capacity utilization path, and Panel B tracks the historical trajectory of the exploitation rate. This long-run alignment illustrates that the surviving multi-equation system embeds a persistent connection between capacity utilization and explicit distributional conditioning.

Figure 7: S2 Focal Utilization and Exploitation Diptych



Source: Author's elaboration.

Exploitation 1965 > 1985 ???

Stage S2 establishes the most demanding empirical restriction of the replication architecture. The traditional output–capital relation fails as an independent bivariate system; its long-run stability depends entirely on the explicit integration of the logged rate of exploitation. The final synthesis indicates that while the technical production relation remains historically trackable, its parametric expression is non-unique, highly sensitive to specific historical controls, and fundamentally dependent on distributional conflict.

4.7 Cross-Stage Econometric Synthesis

The empirical findings across the three estimation layers construct a nested narrative of classical-Marxian capacity dynamics. Stage S0 demonstrates the approximate reproducibility of the baseline capacity path, confirming that the initial replication challenge resides in unstated specification choices rather than data degradation. Stage S1 scales this reconstruction into an admissible space of disciplined non-uniqueness, where the long-run parameter varies continuously along a Pareto fit–complexity frontier. Stage S2 imposes joint multi-equation validation, revealing that the output–capital relation fractures as a standalone entity and achieves empirical stability exclusively when conditioned on the logged rate of exploitation and explicit historical shock vectors.

Table 11: Cross-Stage Synthesis of the Cointegration Architecture

Stage	Outcome	Interpretive Bound
S0	Approximate single-equation reproducibility	Shaikh's method is reproducible, but not exactly
S1	Non-empty admissible ARDL grid; non-unique retained benchmark	Model selection is driven by information-criterion choice
S2 Bivariate	36 estimated, 0 admissible	Output–capital system is not self-sufficient
S2 Trivariate Rank One	36 estimated, 6 admissible	Restricted survival with logged exploitation and h_2 controls
S2 Trivariate Rank Two	36 estimated, 0 admissible	No retained dual-vector system

Source: Author's elaboration.

The magnitude of the estimated transformation elasticity ($\hat{\theta}$) dictates both the historical baseline of the capacity utilization path and its dynamic mean-reverting properties. Under Shaikh's ~~methodology,~~ utilization is constructed strictly as the residual distance between realized output and fitted capacity ($\ln \hat{\mu}_t = y_t - \hat{y}_t^p$). Consequently, $\hat{\theta}$ determines exactly how much of the smooth, long-run capital trend is subtracted from volatile current output. Lower parameter estimates force the residual to absorb structural drift, generating a utilization path characterized by long historical waves and dampened mean reversion. Conversely, higher parameter estimates aggressively strip the capital trend from the residual, resulting in a highly stationary trajectory that mirrors short-run business cycle volatility.

Multi-equation vector estimation resolves this single-equation ambiguity by exposing the historical and institutional conditions necessary for structural stability. The six surviving trivariate VECM specifications strictly require the inclusion of the h_2 historical dummy vector, which accounts for the structural breaks of 1956, 1974, and 1980. This uniform survival pattern demonstrates that the outlier controls implemented in Shaikh's baseline operate as systemic prerequisites for capturing a cointegrated space, rather than arbitrary statistical adjustments. Cointegration is rescued only when these institutional shock vectors stabilize the long-run correlations between accumulation and production.

The structural failure of the standalone output–capital relation stems from the dual macroeconomic role of capital accumulation. Because investment demand expands current output while the capital stock simultaneously builds productive capacity, the scalar parameter $\hat{\theta}$ is forced to mediate conflicting short-run and long-run forces. In a single-equation framework, this accounting entanglement creates severe omitted variable bias and parameter instability. The Stage S2 trivariate system demonstrates that the empirical measurement of capacity utilization cannot be resolved as a purely technical engineering problem. Productive capacity is fundamentally a political economy

??

174 → 80 ok. 56?

the key issue is how latent capacity is estimated

what variables?

In what sense is capacity P.E? Isn't capacity utilization the PE variable?

object, trackable only when the estimation framework explicitly integrates the distributional conflict captured by the logged rate of exploitation.

5 Conclusion

Shaikh shifted the measurement of capacity utilization away from institutional survey conventions and grounded it within the accounting architecture of profitability. This approach correctly recognizes that productive slack cannot be evaluated independently of capital accumulation. However, because the normal rate of profit relies entirely on the constructed capacity denominator, any specification error within the output–capital relation induces a cascading distortion throughout the broader theory of crisis. The critical replication demonstrates that while the single-equation capacity path is arithmetically reproducible, its foundational parameter cannot bear the structural weight assigned to it. The fixed-parameter closure forces a time-invariant multiplier to absorb shifting distributive regimes, generating a highly fragile empirical benchmark.

reference the equations

The instability of this single-equation benchmark exposes a ~~deep theoretical~~ conflict between the accounting mechanics of accumulation and the assumption of proportional reproduction. Interpreting the output–capital coefficient as a transformation elasticity (θ) under an unbalanced growth closure ($\theta \neq 1$) reveals the contradictory macroeconomic role of capital accumulation. Investment demand expands current realized output through the short-run multiplier process. Simultaneously, the expanding capital stock alters the long-run productive ceiling through the transformation elasticity.

When capital accumulation outpaces capacity formation ($\theta < 1$), the macroeconomic system develops a structural drift toward overaccumulation. The empirical system avoids immediate explosive divergence along this differential path only because external components of aggregate demand operate as historical stabilizing buffers. State expenditure, worker consumption, and net exports prevent the utilization rate from collapsing under the weight of structural deceleration. The traditional single-equation specification misreads this bounded historical trajectory as a purely technical, mean-reverting engineering law.

This identification failure confirms that constant algebraic accounting identities cannot substitute for behavioral laws of production. Estimating a cointegrating vector between a macroeconomic flow and a stock without parameterizing the institutional environment treats the historical organization of capital as a passive backdrop. The multi-equation validation in Stage S2 proves that systemic cointegration is achieved exclusively when the logged rate of exploitation anchors the state vector alongside explicit outlier controls for historical structural breaks. Distribution therefore operates as a mathematical prerequisite for the existence of the long-run equilibrium, rather than a secondary interpretive layer.

The objective of this critical replication is to establish the absolute boundary conditions for structural capacity measurement. Identifying the long-run path of productive capacity remains essential for diagnosing the trajectory of capitalist accumulation. Within this initial empirical evaluation, the fixed transformation elasticity (θ) has functioned strictly as a theoretical placeholder to expose the limits of single-equation identification. To construct a valid macroeconomic architecture moving forward, this parameter can no longer be treated as an exogenous scalar. It must be formally endogenized as a dynamic function of distributional conflict and historical regime shifts.

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$\frac{d\hat{k}}{dt}$ } 2nd deriv?

A Derivation of Accumulation Dynamics under Unbalanced Growth

This appendix provides the derivation of the dynamical equations summarized in §?? and §3.4. All results follow from three components: the accounting identity $\hat{k} + \delta = \phi^p B$ (equation ??), the unbalanced growth closure $\hat{y}^p = \theta(\Lambda)\hat{k}$ (equation ??), and the investment-savings closure $\phi^p = s$ (equation ??).

A.1 The Unbalanced Growth Closure: Derivation of the Bernoulli ODE

The accounting identity under the investment-savings closure is:

$$\hat{k} + \delta = s\,B. \tag{23}$$

where B...

Differentiating with respect to time yields:

$$\frac{d\hat{k}}{dt} = s\,\dot{B}. \tag{24}$$

Expressing $\dot{B}$ in terms of the growth rate of capital productivity ($\hat{b}$):

$$\dot{B} = B\,\hat{b}. \tag{25}$$

Substituting this into equation (24) and utilizing the accounting identity $sB = \hat{k} + \delta$ results in the acceleration identity:

$$\frac{d\hat{k}}{dt} = (\hat{k} + \delta)\,\hat{b}. \tag{26}$$

The unbalanced growth closure $\hat{b} = (\theta - 1)\,\hat{k}$ provides the acceleration dynamic:

$$\boxed{\frac{d\hat{k}}{dt} = (\theta - 1)\,\hat{k}\,(\hat{k} + \delta)}. \tag{27}$$

This is a Bernoulli equation of order 2 in $\hat{k}$, where the nonlinearity arises from the product $\hat{k}(\hat{k} + \delta)$.

A.2 Analytical Solution

Expanding equation (27) gives:

$$\frac{d\hat{k}}{dt} = (\theta - 1)\,\hat{k}^2 + (\theta - 1)\,\delta\,\hat{k}. \tag{28}$$

Applying the Bernoulli substitution $v \equiv \hat{k}^{-1}$ with $\dot{v} = -\hat{k}^{-2}\dot{\hat{k}}$ transforms the equation into a first-order linear ODE:

$$\dot{v} + (\theta - 1)\delta\,v = -(\theta - 1). \tag{29}$$

The solution for $v(t)$, given initial conditions $v_0 = \hat{k}_0^{-1}$, is:

$$v(t) = \left(v_0 + \frac{1}{\delta}\right) e^{-(\theta-1)\delta\,t} - \frac{1}{\delta}. \tag{30}$$

Inverting $v(t)$ yields the explicit path for the net accumulation rate:

$$\boxed{\hat{k}(t) = \frac{1}{\left(\frac{1}{\hat{k}_0} + \frac{1}{\delta}\right) e^{-(\theta-1)\delta t} - \frac{1}{\delta}}} \tag{31}$$

A.3 Equilibria and Stability Analysis

The equilibria $\hat{k}^*$ are defined by $(\theta - 1)\,\hat{k}\,(\hat{k} + \delta) = 0$. For $\theta \neq 1$, the roots are $\hat{k}_1^* = 0$ (stationary net capital) and $\hat{k}_2^* = -\delta$ (zero gross investment). Linearizing around $\hat{k}^* = 0$ via the Jacobian $f'(\hat{k}) = (\theta - 1)(2\hat{k} + \delta)$ yields:

- Over-accumulation ($\theta < 1$): $f'(0) < 0$ — the stagnation equilibrium is locally stable.
- Under-accumulation ($\theta > 1$): $f'(0) > 0$ — the stagnation equilibrium is unstable.

A.4 The Trend-Stabilized Closure: Derivation of the Quadratic ODE

Under the trend-stabilized closure $\hat{y}^p = \theta\hat{k} + b$, the growth rate of capital productivity is:

$$\hat{b} = (\theta - 1)\hat{k} + b. \tag{32}$$

Substituting this into the acceleration identity (equation 26) generates a quadratic ODE:

$$\frac{d\hat{k}}{dt} = (\hat{k} + \delta)\left[(\theta - 1)\hat{k} + b\right]. \tag{33}$$

The equilibria are the roots of $(\hat{k} + \delta)[(\theta - 1)\hat{k} + b] = 0$:

$$\hat{k}_1^* = -\delta, \qquad \hat{k}_2^* = \frac{b}{1 - \theta}. \tag{34}$$

The Jacobian $g'(\hat{k}) = 2(\theta - 1)\hat{k} + (\theta - 1)\delta + b$ evaluated at $\hat{k}_2^*$ confirms local stability for the empirically relevant case ($\theta < 1$ and $b > 0$): the interior root $b/(1 - \theta)$ acts as a strictly positive attractor, preventing stagnation. As $b \to 0$, the trend-stabilized closure collapses to the unbalanced growth attractor at $\hat{k}^* = 0$.

B Data Construction and Diagnostic Support

B.1 Series Glossary

Table 12 lists the core replication variables, their sources, and transformation rules. All series derive from Shaikh's (2016) data companions via Real Economic Analysis.[3]

Table 12: Replication series: symbols, sources, and transformations

Symbol / Code	Variable	Source (table, line)	Transformation
GVAcorp_t	Corporate gross value added (corrected)	NIPA 1.14 (L1) + imputed-interest adj. (Table 7.11)	nominal
KGCcorp_t	Corporate gross capital stock (GPIM-adjusted)	Fixed Asset 6.1 (L9); IRS Statistics of Income; GPIM recursion	nominal
KNCcorp_t	Corporate net capital stock (GPIM-adjusted)	same as above	nominal
KNCcorpbea_t	Corporate net capital stock (BEA 2011 official)	Fixed Asset 6.1 (L9)	nominal
p_t	Common deflator	pKN index, 2005=100	(KNCcorp/KNRcorp)*100
y_t	Real log output	constructed	$\ln(\text{VAcorp}_t/p_t)$
k_t	Real log capital	constructed	$\ln(\text{KGCcorp}_t/p_t)$
Profshcorp_t	Corporate profit share (corrected)	NIPA 1.14 (L8, L1−L2) + imputed-interest adj.	ratio

Note: All BEA and NIPA data downloaded in the replication vintage. Shaikh's original series used data downloaded in 2011 (Shaikh, 2016, Appendix 6.7, fn. 1); discrepancies attributable to subsequent BEA revisions are possible and noted where material.

Source: Real Economic Analysis Data Tables

B.2 Imputed-Interest Correction

The NIPA corporate value added figure includes imputed financial intermediation services as a cost to non-financial corporations. The corporate imputed-interest adjustment removes this component:

$$\text{CorpImpIntAdj}_t \equiv -\text{BankMonIntPaid}_t - \text{CorpNFNetImpIntPaid}_t, \tag{35}$$

where BankMonIntPaid_t is NIPA Table 7.11 (lines 4+44+73 minus lines 28+52+91) and CorpNFNetImpIntPaid is NIPA Table 7.11 (line 74 minus line 53). In 2009 the adjustment raises corporate net operating surplus by approximately 10 %. Figure 8 displays the corrected versus uncorrected series and their ratio.

B.3 Corrected Corporate Output: GVAcorp_t

$$\text{GVAcorp}_t \equiv \text{GVAcorpnipa}_t + \text{CorpImpIntAdj}_t, \tag{36}$$

where GVAcorpnipa_t is NIPA Table 1.14 (line 1) and CorpImpIntAdj_t is defined in equation (35).

[3]Replication data and construction scripts are available at Real Economic Analysis: `https://www.realecon.org/data`.

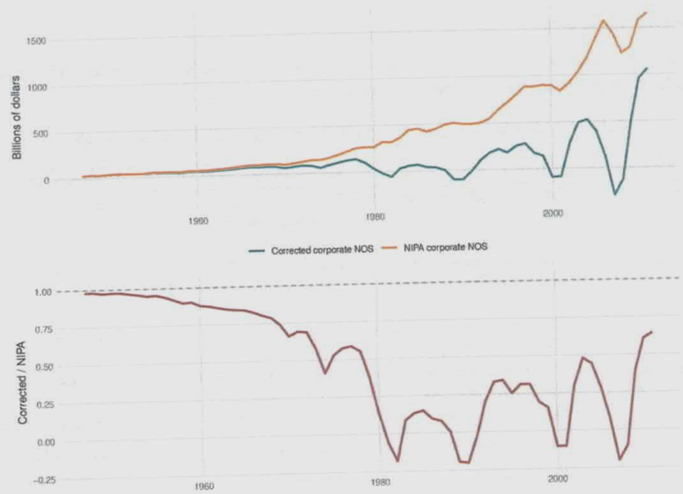

Figure 8: Imputed interest adjustment impact: corrected corporate NOS versus NIPA corporate NOS (1947–2011). Ratio shows magnitude of adjustment.

B.4 Reconstructed Corporate Capital Stock: KGCcorp$_t$

GPIM accumulation rule. The current-cost capital stock is accumulated as:

$$K_t = IG_t + z_t^* \cdot K_{t-1}, \qquad z_t^* \equiv (1 - z_t)\frac{p_t^K}{p_{t-1}^K}, \tag{37}$$

where IG_t is nominal gross investment, z_t is the period depreciation rate, and z_t^* is the survival-and-revaluation factor that reflates the surviving stock to current replacement cost.

Three adjustments (applied before the recursion).

1. **Depletion rates.** BEA 1993 finite service lives replace the BEA 2011 infinite-lives assumption, implying higher depreciation and lower retirement rates consistent with physical exhaustion.
2. **Initial values.** BEA 1993 initial values anchored to 1925 replace the BEA 2011 starting values. The 1993 values are lower, producing faster accumulation growth and convergence to the official path by approximately 1977.
3. **Great Depression correction.** The IRS corporate book value index (of the Census, 1975, Series V 115, pp. 924–926) is applied over 1925–1947, rescaling the BEA historical stock by the IRS/BEA ratio each year. The GPIM recursion resumes from 1947. Net effect: KNCcorp$_t$ is approximately 28 % lower in 1947 than the BEA 2011 official measure.

Inventories. IRS corporate balance sheet inventory ratios are extrapolated over 1947–1989 via a linear fit to the 1990–2011 IRS/BEA ratio, rescaled to the adjusted GPIM stocks, and added to KGCcorp$_t$. Source: IRS Statistics of Income 1926–2011; FRB Flow of Funds FL105015205.A for non-financial corporate inventories in current cost.

Accuracy. The average ratio of the GPIM approximation to the BEA current-cost stock over 1947–2005 is 99.60 %.

why improving? ✓

Figure 9: GPIM-adjusted net capital stock versus BEA 2011 official series (1947–2011). Ratio plot shows convergence by 1977.

B.5 Common Deflator: p_t

The replication uses the pKN implicit price deflator based at $100 = 2005$, estimated under GPIM rules. Current BEA releases employ quality-adjusted chain weights that adjust for quality vintage, breaking stock-flow consistency for accumulation accounting. The pKN index preserves stock-flow consistency, ensuring the output–capital relation reflects real productive dynamics rather than quality-adjustment artifacts (Shaikh, 2016, Appendix 6.6, Sec. II).

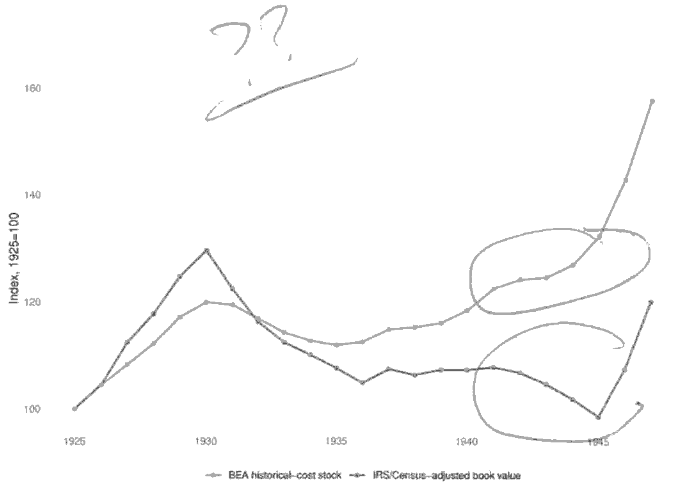

Figure 10: IRS corporate book value index versus BEA historical stock (1925–1947). Both indexed to 100 in 1925.

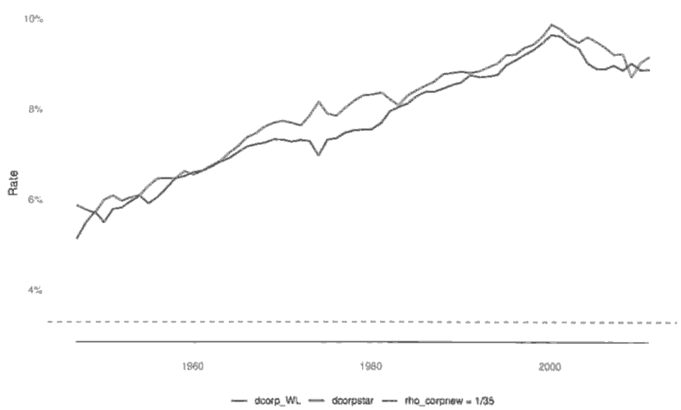

Figure 11: Depreciation and retirement rates under BEA 1993 finite service lives. Horizontal line marks critical value (3.29%).

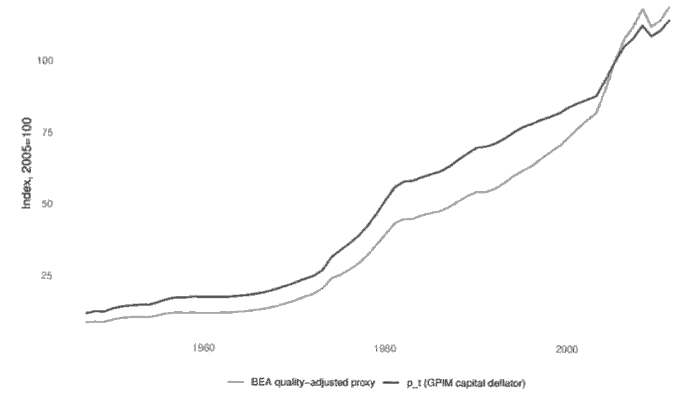

Figure 12: Common deflator index: GPIM-consistent pKN versus BEA quality-adjusted proxy (2005=100).

B.6 Corrected Corporate Profit Share: Profshcorp$_t$

The same imputed-interest adjustment applied to GVAcorp$_t$ is applied to both the numerator and denominator of the profit share:

$$\mathrm{NOScorp}_t \equiv \mathrm{NOScorpnipa}_t + \mathrm{CorpImpIntAdj}_t, \tag{38}$$

$$\mathrm{VAcorp}_t \equiv \mathrm{VAcorpnipa}_t + \mathrm{CorpImpIntAdj}_t, \tag{39}$$

$$\mathrm{Profshcorp}_t \equiv \frac{\mathrm{NOScorp}_t}{\mathrm{VAcorp}_t}, \tag{40}$$

where NOScorpnipa$_t$ is NIPA Table 1.14 (line 8) and VAcorpnipa$_t$ is NIPA Table 1.14 (line 1 minus line 2).

B.7 Descriptive Statistics and Unit Root Tests

Table 13 reports summary statistics for the core series. Table 14 presents unit root test results (ADF, Phillips–Perron) for y_t and k_t in levels and first differences.

Table 13: Core series summary statistics (1947–2011)

Variable	N	Mean	Median	Std Dev	Min	Max	Skewness	Kurtosis
y_t	65	14.9281	14.8653	0.5674	13.8866	15.7628	−0.1438	1.8398
k_t	65	15.9658	15.9810	0.7313	14.7979	17.1127	−0.0206	1.6872
Δy_t	64	0.0293	0.0355	0.0391	−0.0979	0.1055	−0.6331	3.4760
Δk_t	64	0.0362	0.0361	0.0078	0.0186	0.0530	−0.1127	2.4022

p_t	65	50.3486	46.8900	32.8159	11.8262	114.5539	0.3955	1.8063
D_{56}	65	0.0154	0.0000	0.1240	0.0000	1.0000	7.8750	63.0156
D_{74}	65	0.0154	0.0000	0.1240	0.0000	1.0000	7.8750	63.0156
D_{80}	65	0.0154	0.0000	0.1240	0.0000	1.0000	7.8750	63.0156

Source: Real Economic Analysis (`https://www.realecon.org/data`), Shaikh (2016) data companions. Variables constructed per Appendix 12. Dummy variables are impulse indicators with mean $1/65 \approx 0.0154$.

Table 14: Unit root tests for core series (1947–2011)

Variable	Form	Test	Specification	Statistic	Lag/BW	Critical values			Reject
						1%	5%	10%	at 5%
y_t	level	ADF	none	3.0027	1	−2.60	−1.95	−1.61	no
y_t	level	PP	intercept	−1.3280	3	−3.53	−2.91	−2.59	no
y_t	level	PP	trend	−2.1341	3	−4.11	−3.48	−3.17	no
y_t	Δ	ADF	intercept	−5.0966	1	−3.51	−2.89	−2.58	yes
y_t	Δ	PP	intercept	−5.9637	3	−3.54	−2.91	−2.59	yes
y_t	Δ	PP	trend	−5.9460	3	−4.11	−3.48	−3.17	yes
k_t	level	ADF	none	1.1552	4	−2.60	−1.95	−1.61	no
k_t	level	PP	intercept	−0.3863	3	−3.53	−2.91	−2.59	no
k_t	level	PP	trend	−1.2114	3	−4.11	−3.48	−3.17	no
k_t	Δ	ADF	intercept	−2.0670	3	−3.51	−2.89	−2.58	no
k_t	Δ	PP	intercept	−1.9027	3	−3.54	−2.91	−2.59	no
k_t	Δ	PP	trend	−1.8431	3	−4.11	−3.48	−3.17	no

ADF: Augmented Dickey–Fuller test with max lag 6 and AIC selection. PP: Phillips–Perron test with Newey–West bandwidth. "none" = no deterministic terms; "intercept" = constant only; "trend" = constant + linear trend. PP "none" specification not computed. Rejection at 5% indicates stationarity under that specification.

B.8 ARDL Specification Search and Bounds Testing

Table 15 displays the full specification grid (p, q = 1–6; Cases I–V) with information criteria, diagnostic tests, and recovered long-run coefficients. Asterisks mark case-specific AIC/BIC minima.

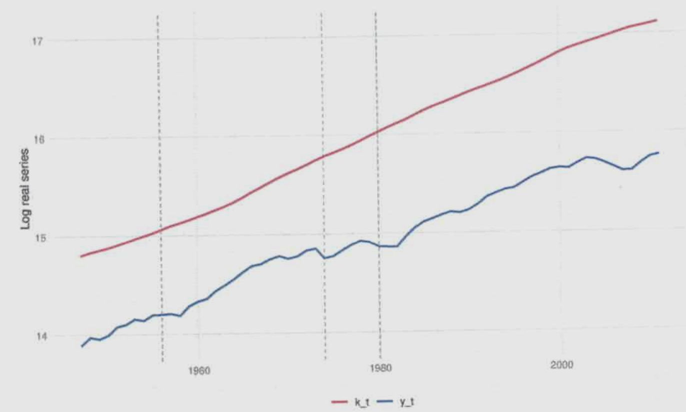

Figure 13: Log output (y_t) and log capital (k_t), 1947–2011. Vertical dashed lines mark 1956, 1974, and 1980.

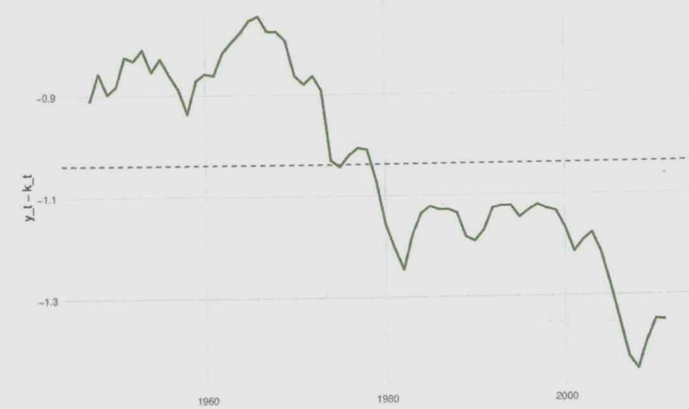

Figure 14: Output–capital ratio ($y_t - k_t$), 1947–2011. Horizontal line marks long-run mean.

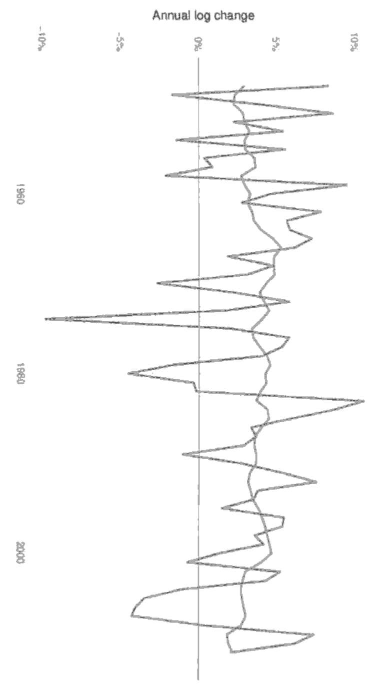

Figure 15: Annual growth rates of output (Δy_t) and capital (Δk_t), 1947–2011.

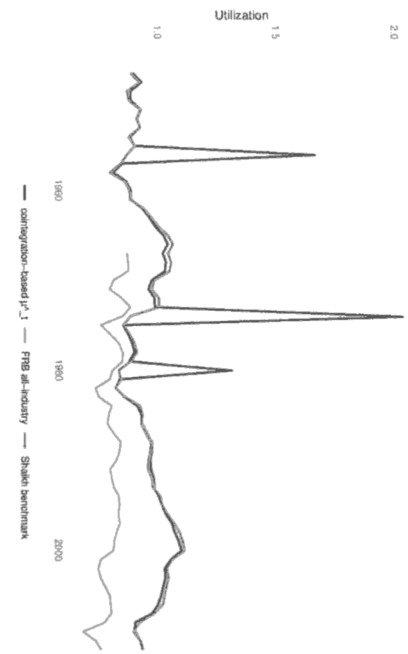

Figure 16: Capacity utilization comparison: Shaikh (cointegration-based) versus FRB survey measure, 1947–2011. Descriptive comparison only.

Table 15: ARDL specification search: information criteria and diagnostics (1947–2011)

p Best	q	Case	AIC	SBC	T_{eff}	$\hat{\theta}$	LM(1)	p-val	LM(4)	p-val
1	1	I	−237.63	−222.52	64	0.8529	3.2749	0.070	5.0893	0.278
1	1	II	−243.09	−225.82	64	0.6989	3.7150	0.054	6.9003	0.141
1	1	III	−243.09	−225.82	64	0.6989	3.7150	0.054	6.9003	0.141
1	1	IV	−243.09	−223.66	64	−0.5048	3.4486	0.063	6.5385	0.162
1	1	V	−243.09	−223.66	64	−0.5048	3.4486	0.063	6.5385	0.162
1	2	I	−239.33	−222.19	63	0.7273	7.0198	0.008	11.1845	0.025
1	2	II	−242.69	−223.41	63	0.7110	7.1117	0.008	11.8692	0.018
1	2	III	−242.69	−223.41	63	0.7110	7.1117	0.008	11.8692	0.018
1	2	IV	−241.90	−220.47	63	−0.4718	7.0457	0.008	11.2449	0.024
1	2	V	−241.90	−220.47	63	−0.4718	7.0457	0.008	11.2449	0.024
1	3	I	−244.02	−224.88	62	0.8967	11.3499	0.001	14.1599	0.007
1	3	II	−249.33	−228.05	62	0.6695	10.3605	0.001	12.7048	0.013
1	3	III	−249.33	−228.05	62	0.6695	10.3605	0.001	12.7048	0.013
1	3	IV	−249.62	−226.22	62	−0.8749	10.3560	0.001	11.9612	0.018

*										
1	3	V	−249.62	−226.22	62	−0.8749	10.3560	0.001	11.9612	0.018
*										
2	3	I	−250.91	−229.63	62	0.9301	5.4621	0.019	9.3503	0.053
2	3	II	−262.79	−239.40	62	0.7023	0.5641	0.453	3.5633	0.468
*										
2	3	III	−262.79	−239.40	62	0.7023	0.5641	0.453	3.5633	0.468
*										
2	3	IV	−264.33	−238.81	62	−0.4836	0.2849	0.594	2.6102	0.625
*										
2	3	V	−264.33	−238.81	62	−0.4836	0.2849	0.594	2.6102	0.625
*										
4	5	I	−253.33	−224.01	60	0.7899	0.0461	0.830	7.6617	0.105
*										
4	5	II	−257.75	−226.33	60	0.7102	0.2628	0.608	6.6363	0.156
4	5	III	−257.75	−226.33	60	0.7102	0.2628	0.608	6.6363	0.156
4	5	IV	−257.11	−223.60	60	−0.4117	0.2120	0.645	6.5141	0.164
4	5	V	−257.11	−223.60	60	−0.4117	0.2120	0.645	6.5141	0.164

ARDL(p, q) specifications estimated with impulse dummies D_{56}, D_{74}, D_{80}. AIC/SBC: Akaike/Schwarz information criteria (lower = better). T_{eff}: effective sample size after lag construction. $\hat{\theta}$: long-run output–capital coefficient recovered from ECM parameters. LM(1)/LM(4): Breusch–Godfrey serial correlation tests (null = no autocorrelation). Asterisk (*) marks best specification by AIC/SBC within each deterministic case. Full grid (p, q = 1–6; Cases I–V) available in replication code.

Figure 17: F-statistic across deterministic Cases I–V. Horizontal lines mark lower ($I(0)$) and upper ($I(1)$) critical bounds at 5% significance.

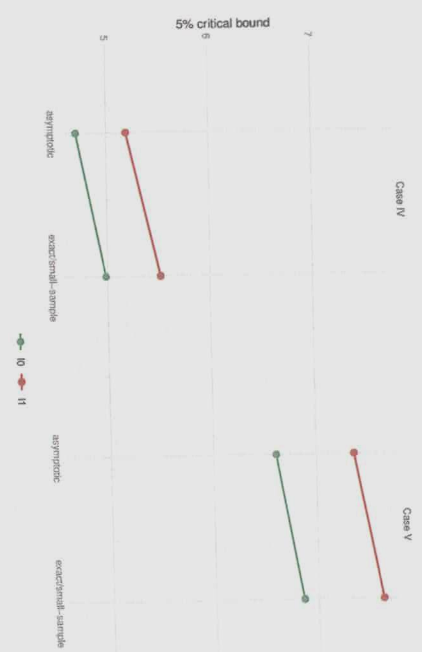

Figure 18: Exact-sample versus asymptotic critical bounds for Cases IV and V ($T = 65$). Exact bounds computed via stochastic simulation (Natsiopoulos and Tzeremes, 2022).

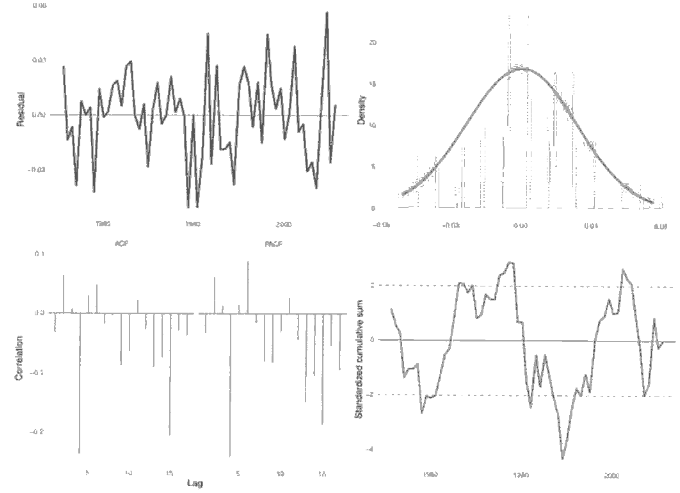

Figure 19: Residual diagnostics from baseline ARDL(2,4): (a) residuals over time, (b) density with normal curve, (c) ACF/PACF, (d) CUSUM test.

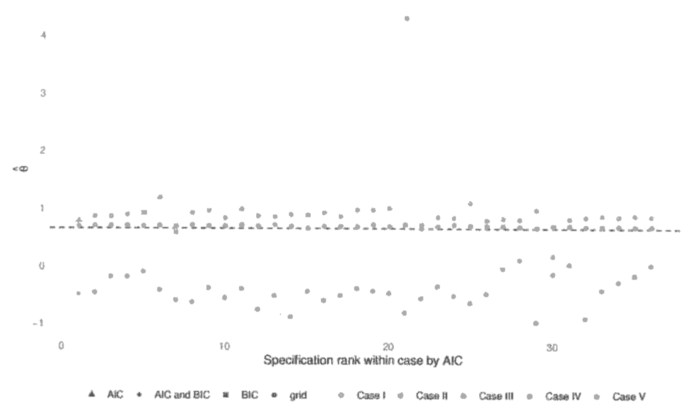

Figure 20: Long-run coefficient $\hat{\theta}$ estimates ranked by AIC across specification grid. Horizontal band marks Shaikh's 0.6609 ± 5%.

B.9 Dummy Variable Diagnostics

Figure 21 displays squared residuals from the baseline ARDL specification with impulse dummies D_{56}, D_{74}, D_{80} overlaid. The dummies absorb outlier-driven residual spikes identified via squared-error diagnostics (Patterson, 2000).

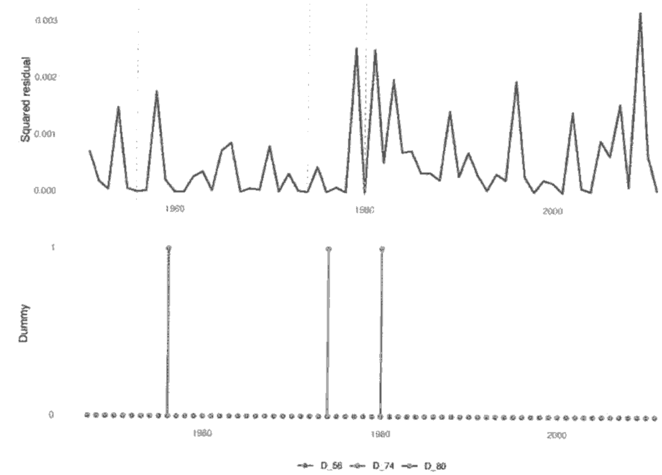

Figure 21: Squared residuals from baseline ARDL with impulse dummies D_{56}, D_{74}, D_{80} (1947–2011). Dummies absorb outlier-driven spikes.

B.10 Series Construction Protocol

Figure 22 visualizes the pipeline from raw data to fitted capacity utilization ($\hat{\mu}_t$). The flowchart traces corrections, deflation, GPIM recursion, ARDL estimation, and residualization.

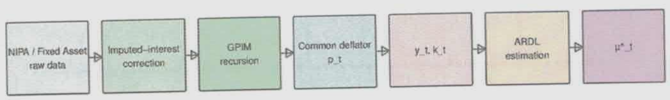

Figure 22: Series construction protocol: raw data $\to$ corrected output/GPIM capital $\to$ common deflator $\to$ ARDL estimation $\to$ $\hat{\mu}_t$.